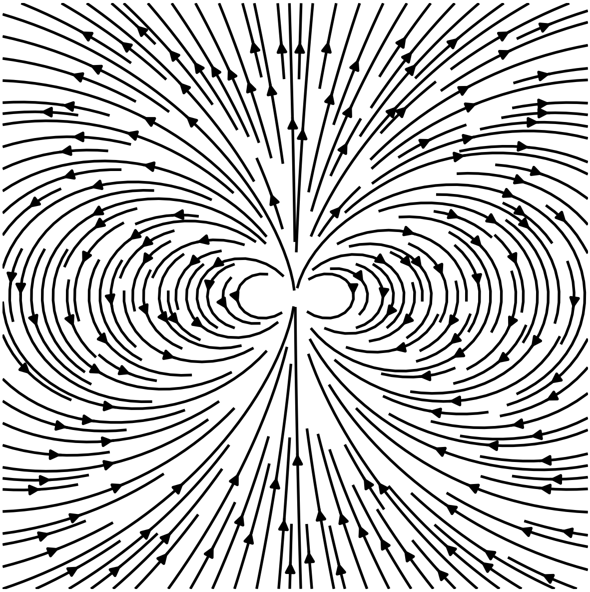


Lecture Notes on Electromagnetic Theory

Physics 131

Vercil Juan
IV - BS Physics

1st Semester, AY 2026-2027

$$
\begin{aligned}
\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0} \\
\nabla \cdot \mathbf{B} &= 0 \\
\nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\
\nabla \times \mathbf{B} &= \mu_0 \mathbf{j} + \frac{1}{c^2}\frac{\partial \mathbf{E}}{\partial t}
\end{aligned}
$$

Dixitque DEUS*: Fiat Lux. Et facta est lux.*

Physics 131 - Electromagnetic Theory I
Course Information

Course Description

Description: Electrostatics in a vacuum, electrostatics in dielectric media, boundary value methods in electrostatics, electric currents, conducting media, magnetostatics in a vacuum, macroscopic and microscopic magnetism, Faraday's law of electromag- netic induction.

Credits: 3.0

Instructor Details

Instructor: Dr. Francis N. C. Paraan

E-mail: `fparaan.upd.edu.ph`

Lecture Hours: WF, 8:30 - 10:00 AM

Course Requirements

You must submit two (2) notebook files, one (1) problem set, and take one sit-in midterm exam to complete the course. Notebooks and problem sets may be handwritten or typeset, but the submission must be a PDF file to be submitted through an online form. The submission link will be provided on the course webpage. You must submit your work by the deadline, otherwise the work will be considered a non-submission. The submission bin will close 48 hours after the deadline as a grace period, and late submissions within this period must be accompanied by a University-recognized excuse. On-site attendance will be checked to comply with University attendance requirements.

Progress will be monitored by performing two types of checkpoint activities:

1. Notebook Entries
2. Problem Sets

Main References

The Syllabus for the course can be found here:

- Course Syllabus

The main reference for this course is:

- David J. Griffiths, Introduction to Electrodynamics 4th Edition (2013)

Note :

The hyperlinks in this lecture notes act as follows: The ■ hyperlink, when pressed, opens the relevant `pdf` in my (the Author's) local machine. This only works for my personal computer. The hyperlink on the **name of the book** links to a source of the book on the internet. Press that hyperlink to access that book.

Reminders

Exam Dates

- ☐ Midterm Exam :

Problem Set Dates

- ☐

Notebook Submission Dates

- ☐
- ☐

Preface

These notes present a systematic summary of the material developed in Physics 131, Electromagnetic Theory I, with David J. Griffiths' *Introduction to Electrodynamics*, 4th Edition, serving as the principal reference. They are not intended as a verbatim transcription of lectures–though the author does indeed copy the text verbatim in some sections–or as a replacement for the primary text, but rather as a consolidated reference in which the principal definitions, mathematical methods, physical arguments, and results have been selected and reorganized for review. The presentation also incorporates relevant material from other parts of the undergraduate physics and mathematics curriculum, particularly vector calculus and mathematical methods encountered in Physics 116, where such material is useful for establishing the mathematical framework required for electromagnetism. Relevant graphs were made using Python and `matplotlib`, with additional material such as relevant problem sets added.

The organization follows the development of the subject from the mathematical foundations of vector calculus through electrostatics and magnetostatics, with emphasis on the connection between the mathematical description of fields and their physical interpretation. The material begins with vectors, differential and integral calculus, curvilinear coordinates, and the fundamental theorems of vector calculus before developing Coulomb's law, electric fields, magnetic fields, and other associated topics. These notes were prepared primarily as a personal reference for the author's own study and review, although they have been organized sufficiently to serve as a supplementary reference for classmates, peers, and other students who may find them useful.

These lecture notes was prepared by the author using $\LaTeX\,2_\varepsilon$.

Vercil Juan
V - BS Physics
University of the Philippines Diliman

Physica Vincit Omnia

Contents

List of Figures

List of Theorems

Unit 1

Electostatics

1.1 Review of Vector Calculus

This is basically just a summary of the **Physics 116 curriculum**, and as such, **I will be pasting an abridged copy of my notes from that notebook to here**. Anyway.

1.1.1 Vectors and Scalars

A **scalar** $a \in \mathbb{R}$ is a quantity with magnitude. For the purposes of this notes, scalars are denoted in italic script. A scalar is in the set of real numbers

$$a \in \mathbb{R} \tag{1.1.1}$$

An example of a scalar is the temperature in a room, T_0, time t, and other single valued quantities.

A **vector** is a collection of scalars, usually indicating magnitude and direction. We represent vectors in math bold face. A vector of n dimensions is represented by n numbers, v_i, where i represents the i-th element of the vector. A vector can be a column vector or a row vector.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \qquad \mathbf{v}^\top = \begin{bmatrix} v_1 & v_2 & v_3 \end{bmatrix} \tag{1.1.2}$$

An example of a vector is a particle's position in space

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \tag{1.1.3}$$

A tensor is a generalization of vectors. It is essentially an n dimmensional (More commonly called rank) vector. This is generally beyond the scope of this notes, but I included it for generality. An example of a tensor is the stress energy tensor.

$$T = \begin{bmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{bmatrix} \tag{1.1.4}$$

It's essentially a matrix, but more general. A matrix is a rank 2 tensor. A vector is a rank 1 tensor, while a scalar is a rank 0 tensor.

1.1.1.1 Vector operations

There are several operations that can be done with vectors and scalars. Here are some of them.

Vector addition For a vector $\mathbf{u}$ and $\mathbf{v}$, adding them is done my adding element-wise. The same is true for subtraction.

$$\mathbf{u} \pm \mathbf{v} = (u_i \pm v_i)\hat{\mathbf{e}}_i \tag{1.1.5}$$

Scalar multiplication A vector multiplied by a scalar is equivalent to multiplying all its elements with the scalar. As a shorthand, we can represent vectors and all it's elements with one expression $\mathbf{v} = v_i$, where the index i runs over from 1 to n. Given $a \in \mathbb{R}$, a scalar multiplication is

$$a\mathbf{v} = \begin{bmatrix} av_1 \\ av_2 \\ av_3 \end{bmatrix} = av_i\hat{\mathbf{e}}_i \tag{1.1.6}$$

Dot Product The dot product of two vectors is the sum of the product of the i-th element of each vector. For example, given a vector $\mathbf{v} = v_i$ and $\mathbf{u} = u_i$, their dot product is

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + u_3 v_3 = \sum_i u_i v_i \tag{1.1.7}$$

We will encounter a lot of these sums in general, so we introduce the *Einstein summation convention* where, if we see repeated indices, this implies that the expression is being summed over that index. As such, we can more conveniently define a dot product as

$$\mathbf{u} \cdot \mathbf{v} = u_i v_i \tag{1.1.8}$$

Cross Product The cross product is pretty hard to define, but in general, it is shown the vector which is the determinant of the matrix of the vector elements. But first, we introduce the following vectors

$$\hat{\mathbf{e}}_1 = \hat{\boldsymbol{\imath}} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_2 = \hat{\boldsymbol{\jmath}} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad \hat{\mathbf{e}}_3 = \hat{\boldsymbol{k}} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{1.1.9}$$

There are known as **elementary orthogonal basis vectors** of the vector space. Basically unit vectors of a single dimension. We can represent any vector in $\mathbb{R}^3$ in this notation.

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_1 \hat{\boldsymbol{\imath}} + v_2 \hat{\boldsymbol{\jmath}} + v_3 \hat{\boldsymbol{k}} = v_i \hat{\mathbf{e}}_i \tag{1.1.10}$$

It must be noted that $\mathbf{v} = v_i(x, y, z)\hat{\mathbf{e}}_i$. In that definition, then, we often see the dot product defined as

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = (u_2 v_3 - u_3 v_2)\hat{\boldsymbol{\imath}} - (u_1 v_3 - u_3 v_1)\hat{\boldsymbol{\jmath}} + (u_1 v_2 - u_2 v_1)\hat{\boldsymbol{k}} \tag{1.1.11}$$

The Kronecker Delta and LeviCivita operators The Kronecker delta is an operator which basically turns on or off a product depending on it's indices. Formally, it is defined as

$$\delta_{ij} = \hat{\mathbf{e}}_i \cdot \hat{\mathbf{e}}_j = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases} \tag{1.1.12}$$

and the Levi-Civita symbol is defined to be 0 if two indices are the same, 1 if the indices are an even permutation of i, j, k and -1 if it is not.

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } ijk \text{ is } 123, 312, 231 \\ -1 & \text{if } ijk \text{ is } 132, 321, 213 \\ 0 & \text{if } i = j \text{ or } j = k \text{ or } i = k \end{cases} \tag{1.1.13}$$

with this, we can more succintly define the dot and cross product as

$$\mathbf{u} \cdot \mathbf{v} = \delta_{ij} u_i v_j \tag{1.1.14}$$

$$\mathbf{u} \times \mathbf{v} = \epsilon_{ijk} u_i v_j \hat{\mathbf{e}}_k \tag{1.1.15}$$

Einstein Summation Notation To prevent clunkiness of equations, we oftem omit the summation symbol. By convention, when an index appears **twice** in a single term (once as a superscript and once as a subscript), it is **implicitly summed over** all values of that index.

$$\sum_i a_i b^i \equiv a_i b^i \tag{1.1.16}$$

The summation symbol $\sum$ is ommited. This applies to the tensor notation listend above. Sometimes, it is also not done with superscripts, but the summation convention applies nonetheless.

$$\sum_i a_i b_i \equiv a_i b_i \tag{1.1.17}$$

1.1.2 Vector Properties

1.1.2.1 The Vector Norm

The vector norm is the *length* of the vector, so to speak. It is defined as the square root of the sums of the components, i.e,

$$\|\mathbf{v}\| = \sqrt{\sum_i v_i^2} \tag{1.1.18}$$

In more explicit notation, we have:

$$\|\mathbf{v}\| = \sqrt{\mathbf{v}^2} = \sqrt{\mathbf{v} \cdot \mathbf{v}} \tag{1.1.19}$$

Note that we define the square of a vector to be its dot product with itself. $\mathbf{v}^2 = \mathbf{v}$. Furthermore, sometimes we denote the norm to be italic font of the vector.

$$\|\mathbf{v}\| \equiv v$$

1.1.2.2 Vector Axioms

Technically, a vector is really just defined as an element of a vector space.

$$\mathbf{v} \in \mathbb{V}$$

Vector spaces are an abstract space which obeys the following properties[1].

Definition 1.1 : Vector Axioms

Let $\mathbf{u}$,$\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{V}$. The space $\mathbb{V}$ is a vector space if the following conditions hold:

These operations are defined as follows:

1. Vector addition, $+ : V \times \mathbb{V} \to \mathbb{V}$
2. Scalar multiplication, $\cdot : \mathbb{R} \times \mathbb{V} \to \mathbb{V}$

and these conditions must hold

1. Closure under addition, $\mathbf{u} + \mathbf{v} \in \mathbb{V}$
2. Commutativity, $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3. Associativity, $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{v} + (\mathbf{u} + \mathbf{w})$
4. Additive identity (zero vector $\mathbf{0}$), $\forall \mathbf{v} \in \mathbb{V}, \exists \mathbf{0} \in \mathbb{V} : \mathbf{v} + \mathbf{0} = \mathbf{v}$
5. Additive inverse, $\forall \mathbf{v} \in \mathbb{V}, \exists - \mathbf{v} \in \mathbb{V} : \mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
6. Closure under scalar multiplication, $\forall a \in \mathbb{R} \wedge \mathbf{v} \in \mathbb{V}, a\mathbf{v} \in \mathbb{V}$
7. Distributivity of scalar multiplication under vector addition, $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
8. Distributivity of scalar multiplication under scalar addition, $\forall a, b \in \mathbb{R}, (a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$
9. Associativity of scalar multiplication, $a(b\mathbf{v}) = (ab)\mathbf{v}$
10. Existence of multiplicative identity, $\forall \mathbf{v} \in \mathbb{V}, \exists 1 : 1 \cdot \mathbf{v} = \mathbf{v}$

[1] This comes from the Math 40 curriculum

1.1.2.3 Vector derivatives and Integrals

To take the derivative of a vector, one must apply the derivative on all of it's components, just like normal derivatives. Chain rule and other calculus methods still apply.

$$\frac{d\mathbf{v}}{dt} = \begin{bmatrix} \frac{dv_1}{dt} \\ \frac{dv_2}{dt} \\ \frac{dv_3}{dt} \end{bmatrix} = \frac{dv_i}{dt} \tag{1.1.20}$$

Similarly, the same applies for integration, along with path integrals and the like which will not be part of this section.

$$\int \mathbf{v}\, dt = \begin{bmatrix} \int v_1\, dt \\ \int v_2\, dt \\ \int v_3\, dt \end{bmatrix} = \int v_i\, dt \tag{1.1.21}$$

1.1.3 Vector Transformations

1.1.3.1 Linear transformations

Consider a vector

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = v_i \hat{\mathbf{e}}_i$$

A linear transformation is the operation

$$\mathbf{v}' = A\mathbf{v} \tag{1.1.22}$$

where A is an $n \times n$ matrix. Linearity in this case means

$$A(a\mathbf{u} + b\mathbf{v}) = aA\mathbf{u} + bA\mathbf{v} \tag{1.1.23}$$

The matrix A is described as

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{bmatrix} \tag{1.1.24}$$

Notation A matrix is denoted in index notation as

$$A = a_{ij}$$

It has a few defining properties, but we consider the determinant and the trace. The trace is defined as follows

$$A^\top = a_{ji}$$

It is assumed that the reader already knows the determinant.

Multipliying matrices A and B are defined as follows

$$AB = a_{il}b_{lj}$$

A general 2×2 linear transformation is described in matrix form as

$$\mathbf{v}' = A\mathbf{v} \tag{1.1.25}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} \tag{1.1.26}$$

As such,

$$x' = ax + by \qquad\qquad y' = cx + dy \tag{1.1.27}$$

1.1.3.2 Eigenvectors, Eigenvalues

An **eigenvector** is a vector whose direction is unchanged by a transformation.

Definition 1.2 : Eigenvector

$$A\mathbf{v} = \lambda\mathbf{v} \tag{1.1.28}$$

Where λ is known as the **eigenvalue**. To find an eigenvalue, consider

$$(A - \lambda\mathbb{1})\mathbf{v} = 0 \tag{1.1.29}$$

where $\mathbb{1}$ is an identity matrix

$$\mathbb{1} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} \tag{1.1.30}$$

For a non trivial vector, we have

Definition 1.3 : Characteristic equation

$$\det(A - \lambda \mathbb{1}) = 0 \tag{1.1.31}$$

This is the **characteristic equation**. Once λ is known, (1.1.31) is solvable.

1.1.3.3 Translation

Translation is differernt from ordinary matrix multiplications, because

$$\mathbf{r}' = \mathbf{r} + \mathbf{a} \tag{1.1.32}$$

where $\mathbf{a}$ is another vector. Doing this moves or *translates* the vector.

For example

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} a \\ b \end{bmatrix} \tag{1.1.33}$$

This is not linear, because generally

$$T(a\mathbf{v}) \neq aT(\mathbf{v}) \tag{1.1.34}$$

It is instead an *affine transformation*

A general affine transformation is

$$\mathbf{r}' = A\mathbf{r} + \mathbf{b} \tag{1.1.35}$$

1.1.3.4 Rotation

2 Dimensions A rotation in two dimensions through an angle θ is represented by

Definition 1.4 : Rotation Matrix ($SO(2)$)

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \tag{1.1.36}$$

and, the transformation is defined by

$$\mathbf{v}' = R(\theta)\mathbf{v} \tag{1.1.37}$$

thus

$$x' = x\cos\theta - y\sin\theta \tag{1.1.38}$$
$$y' = x\sin\theta + y\cos\theta \tag{1.1.39}$$

The defining property of a rotation matrix is

$$R^\top R = \mathbb{1} \tag{1.1.40}$$

and

$$\det(R) = \mathbb{1} \tag{1.1.41}$$

The set of all such 2×2 matrices form the special orthogonal group in 2 dimensions $SO(2)$.

Rotations preserve lengths and angles during their transformations.

$$\|\mathbf{v}'\| = \|\mathbf{v}\|$$

3 Dimensions In three dimensions, rotations are represented as much the same.

$$\mathbf{v}' = R\mathbf{v}$$

with the same properties as $SO(2)$. All such matrices form the group known as $SO(3)$. Unlike $SO(2)$, a 3D rotation generally has a rotation axis

$$R_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{1.1.42}$$

$$R_x(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \tag{1.1.43}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \tag{1.1.44}$$

1.1.4 Field

A *field* is a function that assigns a value to every point in a defined space. In other words, a field takes the position of a point in space as its input and returns some physical or mathematical quantity associated with that point.

For a three-dimensional space, we can specify the position of a point using the position vector

$$\mathbf{r} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

The field is then a function of this position vector. Depending on the type of quantity being assigned, we distinguish between scalar fields and vector fields.

1.1.4.1 Scalar field

A scalar field assigns a single scalar value to every point in space. For example, the temperature distribution in a room can be represented by

$$T(\mathbf{r}) = T(x, y, z).$$

Here, the input $\mathbf{r}$ specifies a particular point in space, while the output $T(\mathbf{r})$ is the temperature at that point.

Thus, a scalar field can be thought of as a rule that associates one number with every point:

$$\mathbf{r} \longmapsto T(\mathbf{r}).$$

For example,

$$T(1, 2, 3) = 300 \text{ K}$$

would mean that the temperature at the point $(1, 2, 3)$ is 300 K.

We often write a scalar field as

$$\phi(\mathbf{r}),$$

where ϕ denotes the field and $\mathbf{r}$ specifies the position at which the field is evaluated.

1.1.4.2 Vector field

A vector field assigns a vector to every point in space. For example, the velocity of a fluid flowing through a region of space can be represented by

$$\mathbf{v}(\mathbf{r}) = \mathbf{v}(x, y, z).$$

Unlike a scalar field, which returns a single number, a vector field returns a vector whose components are themselves functions of position.

We can therefore write

$$\mathbf{F}(\mathbf{r}) = \begin{bmatrix} F_1(\mathbf{r}) \\ F_2(\mathbf{r}) \\ F_3(\mathbf{r}) \end{bmatrix} = F_i(\mathbf{r})\hat{\mathbf{e}}_i$$

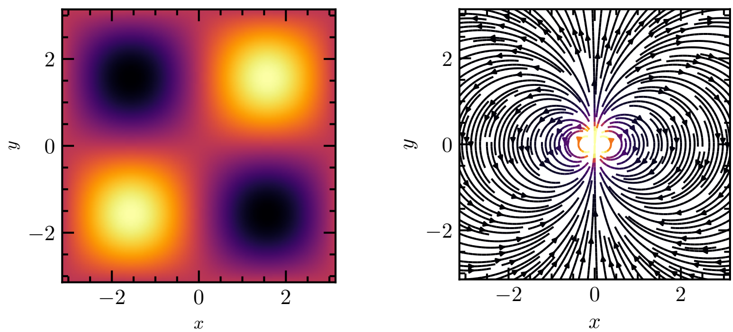
where each component

$$F_i(\mathbf{r})$$

specifies the corresponding component of the vector at the point $\mathbf{r}$.

Thus, at each point in space, the field provides both a direction and a magnitude.

The important distinction is therefore that $\mathbf{r}$ identifies *where* we are evaluating the field, while $\phi(\mathbf{r})$ or $\mathbf{F}(\mathbf{r})$ tells us what *value* the field assigns at that point.



(a) A scalar field (b) A vector field

Figure 1.1: Types of Fields

1.1.5 Differential Calculus

1.1.5.1 The Del Operator

The Del operator ∇ is the generalization of the derivative. We define it as a vector of the partial derivative with respect to the dimension, i.e.

Definition 1.5 : ∇ operator

$$\nabla = \frac{\partial}{\partial x}\hat{\boldsymbol{\imath}} + \frac{\partial}{\partial y}\hat{\boldsymbol{\jmath}} + \frac{\partial}{\partial z}\hat{\boldsymbol{k}} = \begin{bmatrix} \partial/\partial x \\ \partial/\partial y \\ \partial/\partial z \end{bmatrix} \tag{1.1.45}$$

As a shorthand, we can represent it as $\dfrac{\partial}{\partial x_i}$, where x_i is the general dimension. This can be further shorthanded to ∂_i. Thus, a compact vector definition of the Del operator is

$$\nabla = \partial_i \hat{\mathbf{e}}_i \tag{1.1.46}$$

1.1.5.2 Gradient

We often represent fields as $\psi(x, y, z)$; however, we can just make x, y, z into a vector $\mathbf{r} = (x, y, z)$. In that case, the physical interpretation of the gradient becomes clearer.

The **gradient** of a scalar field $\phi(\mathbf{r})$ is a vector field that points in the direction of the greatest rate of increase of ϕ:

Definition 1.6 : Gradient

$$\nabla\phi = \frac{\partial\phi}{\partial x}\hat{\boldsymbol{\imath}} + \frac{\partial\phi}{\partial y}\hat{\boldsymbol{\jmath}} + \frac{\partial\phi}{\partial z}\hat{\boldsymbol{k}} = \begin{bmatrix} \dfrac{\partial\phi}{\partial x} \\ \dfrac{\partial\phi}{\partial y} \\ \dfrac{\partial\phi}{\partial z} \end{bmatrix} \tag{1.1.47}$$

It measures how rapidly the scalar field changes and in which direction this change is maximum. For example, in a temperature field

$T(x, y, z)$, the gradient ∇T points in the direction where temperature increases most rapidly, and its magnitude gives the rate of increase per unit distance.

In index notation, we can write

$$\nabla\phi = \partial_i \phi \, \hat{\mathbf{e}}_i, \tag{1.1.48}$$

where ∂_i denotes differentiation with respect to x_i.

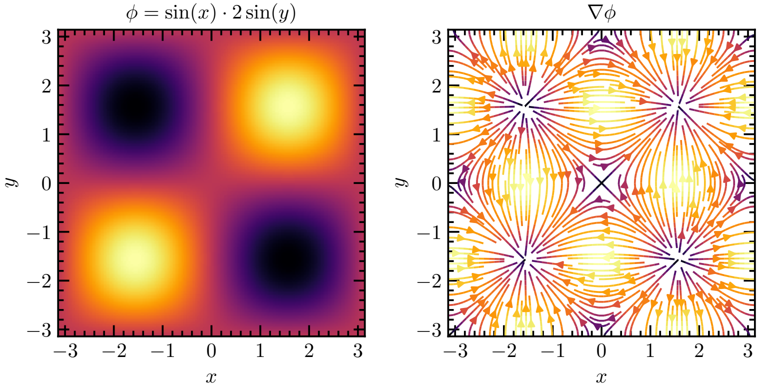


Figure 1.2: A scalar field and its gradient

1.1.5.3 Divergence

The **divergence** of a vector field $\mathbf{F}(\mathbf{r}) = \begin{bmatrix} F_x \\ F_y \\ F_z \end{bmatrix} = F_i \hat{\mathbf{e}}_i$ measures the net flux per unit volume leaving a point:

Definition 1.7 : Divergence

$$\nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}. \tag{1.1.49}$$

A positive divergence indicates a *source* (more field lines leaving than entering), while a negative divergence indicates a *sink* (field lines converging). If $\nabla \cdot \mathbf{F} = 0$, the field is said to be **solenoidal**, meaning it has no net flux out of any closed surface.

This operation is essentially the dot product of the del operator with the vector field:

$$\nabla \cdot \mathbf{F} = \delta_{ij}\ \partial_i F_j, \tag{1.1.50}$$

Divergence appears naturally in continuity equations. For instance, $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ in Gauss's law for electricity, where it represents the electric charge density acting as a source of the field.

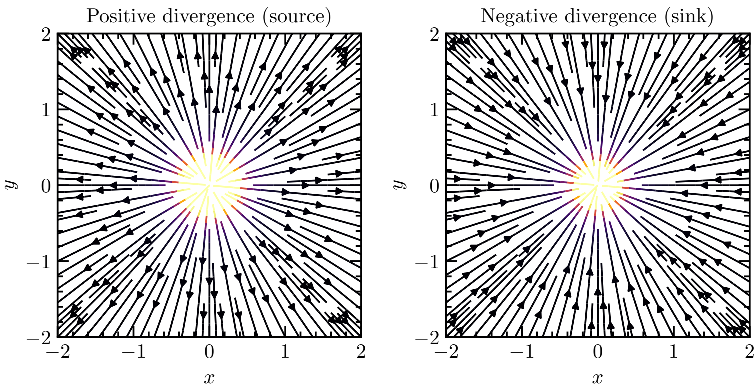


Figure 1.3: Examples of vector fields with positive and negative divergences

1.1.5.4 Curl

The **curl** of a vector field $\mathbf{F}(\mathbf{r})$ measures its local rotation or the tendency to induce circulation around a point:

Definition 1.8 : Curl

$$\nabla \times \mathbf{F} = \begin{vmatrix} \hat{\boldsymbol{\imath}} & \hat{\boldsymbol{\jmath}} & \hat{\boldsymbol{k}} \\ \dfrac{\partial}{\partial x} & \dfrac{\partial}{\partial y} & \dfrac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix}. \tag{1.1.51}$$

The direction of $\nabla \times \mathbf{F}$ gives the axis of rotation (using the right-hand rule), and its magnitude gives the rotational strength.

This is equivalent to taking the cross product of the del operator with the vector field:

$$\nabla \times \mathbf{F} = \varepsilon_{ijk}\, \partial_i F_j \hat{\mathbf{e}}_k \tag{1.1.52}$$

In physical contexts, the curl represents quantities such as the rotational velocity of a fluid or the magnetic field induced by a current. For example, in electromagnetism, $\nabla \times \mathbf{E} = -\,\partial \mathbf{B}/\partial t$ expresses Faraday's law of induction.

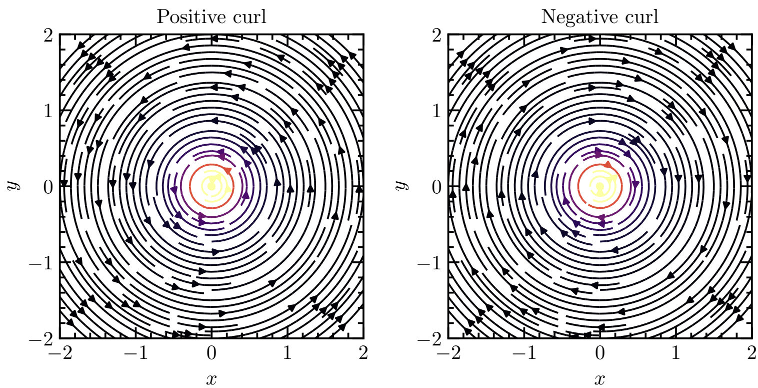


Figure 1.4: Example of vector fields with positive and negative curl

1.1.5.5 Laplacian

The **Laplacian** operator acts on scalar or vector fields and is defined as the divergence of the gradient:

Definition 1.9 : Laplacian

$$\nabla^2\phi = \nabla\cdot(\nabla\phi) = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}. \tag{1.1.53}$$

It measures how a field value at a point compares to its average in the surrounding region — in other words, how “curved” or “peaked” the field is locally. It is worthy to note that the laplacian of a scalar field is also a scalar field.

In index notation, this becomes

$$\nabla^2\phi = \delta_{ij}\ \partial_i\,\partial_j\phi. \tag{1.1.54}$$

The Laplacian appears in many fundamental physical equations:

- Poisson’s equation: $\nabla^2\phi = -\rho/\varepsilon_0$ (electrostatics)
- Heat equation: $\partial\phi/\partial t = \alpha\nabla^2\phi$ (diffusion)
- Newtonian Field equation for Gravity: $\nabla^2\Phi = 4\pi G\rho$

If $\nabla^2\phi = 0$, the field is called **harmonic**, meaning it satisfies Laplace’s equation and represents an equilibrium distribution such as steady-state temperature or potential.

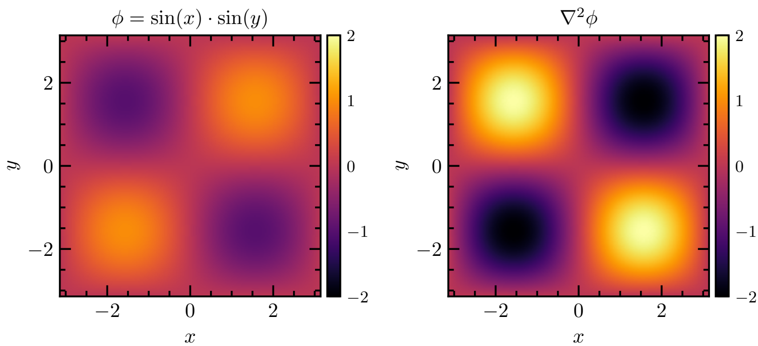


Figure 1.5: Scalar field and its Laplacian

1.1.5.6 Operations on these Operators

The usual differentiation rules still apply for $\nabla, \nabla\cdot$, and $\nabla\times$, because the are built from ordinary partial derivatives. One only has to account for the fact that the fields may be scalar or vector valued.

Product Rule For scalar fields f, g

$$\nabla(fg) = f\,\nabla g + g\,\nabla f \tag{1.1.55}$$

For a scalar f and a vector field $\mathbf{A}$

$$\nabla\cdot(f\mathbf{A}) = f(\nabla\cdot\mathbf{A}) + (\nabla f)\cdot\mathbf{A} \tag{1.1.56}$$

and

$$\nabla\times(f\mathbf{A}) = f(\nabla\times\mathbf{A}) + (\nabla f)\times\mathbf{A} \tag{1.1.57}$$

Chain Rule This also applies for these operations.

Suppose $f = f(u)$ and $u = u(x_1, x_2, x_3)$, then

$$\nabla f(u) = \frac{df}{du}\,\nabla u \tag{1.1.58}$$

More generally, if $f = f(u, v)$ and $u = u(x, y, z)$ and $v = v(x, y, z)$

$$\nabla f(u, v) = \frac{\partial f}{\partial u}\,\nabla u + \frac{\partial f}{\partial v}\,\nabla v \tag{1.1.59}$$

Quotient Rules For scalar fields f, g and vector field $\mathbf{A}$, we have three possible quotient rules.

$$\begin{aligned}
\nabla\left(\frac{f}{g}\right) &= \frac{g\,\nabla f - f\,\nabla g}{g^2}\\
\nabla\cdot\left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla\cdot\mathbf{A}) - \mathbf{A}\cdot(\nabla g)}{g^2}\\
\nabla\times\left(\frac{\mathbf{A}}{g}\right) &= \frac{g(\nabla\times\mathbf{A}) + \mathbf{A}\times(\nabla g)}{g^2}
\end{aligned}$$

Additional Rules Some other useful identities are

$$
\begin{aligned}
\nabla \cdot (\mathbf{A} + \mathbf{B}) &= \nabla \cdot \mathbf{A} + \nabla \cdot \mathbf{B} \\
\nabla \times (\mathbf{A} + \mathbf{B}) &= \nabla \times \mathbf{A} + \nabla \times \mathbf{B}
\end{aligned}
$$

Some useful second-derivative identities are

$$
\begin{aligned}
\nabla \cdot (\nabla f) &= \nabla^2 f \\
\nabla \times (\nabla f) &= \mathbf{0} \\
\nabla \cdot (\nabla \times \mathbf{A}) &= 0 \\
\nabla \times (\nabla \times \mathbf{A}) &= \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}
\end{aligned}
$$

For vector products,

$$
\begin{aligned}
\nabla(\mathbf{A} \cdot \mathbf{B}) &= \mathbf{A} \times (\nabla \times \mathbf{B}) + \mathbf{B} \times (\nabla \times \mathbf{A}) + (\mathbf{A} \cdot \nabla)\mathbf{B} + (\mathbf{B} \cdot \nabla)\mathbf{A} \\
\nabla \cdot (\mathbf{A} \times \mathbf{B}) &= \mathbf{B} \cdot (\nabla \times \mathbf{A}) - \mathbf{A} \cdot (\nabla \times \mathbf{B}) \\
\nabla \times (\mathbf{A} \times \mathbf{B}) &= \mathbf{A}(\nabla \cdot \mathbf{B}) - \mathbf{B}(\nabla \cdot \mathbf{A}) + (\mathbf{B} \cdot \nabla)\mathbf{A} - (\mathbf{A} \cdot \nabla)\mathbf{B}
\end{aligned}
$$

And some useful cross-product and triple product identities are

$$
\begin{aligned}
\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) &= \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\
\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \\
(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} &= \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{A}(\mathbf{B} \cdot \mathbf{C}) \\
\mathbf{A} \times \mathbf{B} &= -\mathbf{B} \times \mathbf{A}
\end{aligned}
$$

1.1.6 Integral Calculus

1.1.6.1 Line integrals

For reference, a line integral is defined as the integral over the path Γ of a parameterized vector path function $\mathbf{r}(t)$ over a scalar field ϕ or a vector field $\mathbf{F}$ with respect to a parameter t. Go figure. It comes in two forms: scalar line integrals and vector line integrals.

For a scalar field, the line integral is

Definition 1.10 : Scalar line Integral

$$\int_\Gamma \phi(\mathbf{r})\ ds = \int_a^b \phi(\mathbf{r}(t))\left\|\frac{d\mathbf{r}}{dt}\right\| dt. \tag{1.1.60}$$

Here, ds is the differential arc length along the curve. Since

$$ds = \left\|\frac{d\mathbf{r}}{dt}\right\| dt, \tag{1.1.61}$$

the scalar line integral can be understood as summing the value of the scalar field along the curve, weighted by the infinitesimal distance traveled along that curve. It is therefore independent of the particular parameterization of the curve, provided that the parameterization preserves its orientation and traces the same path.

For a vector field, the line integral is

Definition 1.11 : Vector line Integral

$$\int_\Gamma \mathbf{F}(\mathbf{r}) \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \frac{d\mathbf{r}}{dt}\, dt. \tag{1.1.62}$$

Unlike the scalar line integral, the vector line integral depends on the direction in which the path is traversed. Reversing the orientation of Γ gives

$$\int_{-\Gamma} \mathbf{F} \cdot d\mathbf{r} = -\int_\Gamma \mathbf{F} \cdot d\mathbf{r}. \tag{1.1.63}$$

The quantity $d\mathbf{r}$ is the infinitesimal displacement vector along the curve,

$$d\mathbf{r} = \frac{d\mathbf{r}}{dt}\,dt, \tag{1.1.64}$$

so the vector line integral measures the component of the vector field along the direction of motion. In physics, this is why the work done by a force $\mathbf{F}$ along a path Γ is written as

$$W = \int_\Gamma \mathbf{F} \cdot d\mathbf{r}. \tag{1.1.65}$$

If the vector field is conservative, so that

$$\mathbf{F} = -\nabla\phi, \tag{1.1.66}$$

then the integral depends only on the endpoints of the curve rather than on the path taken. In that case, we have the fundamental theorem of gradients.

Theorem 1.1 : Fundamental theorem of Gradients

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\mathbf{r} = \phi(\mathbf{a}) - \phi(\mathbf{b}), \tag{1.1.67}$$

where $\mathbf{a}$ and $\mathbf{b}$ are the initial and final points of the path. Equivalently, for contour integral of a closed curve (where the endpoints are the same),

$$\oint_\Gamma \mathbf{F} \cdot d\mathbf{r} = 0. \tag{1.1.68}$$

This gives a useful connection between line integrals and the gradient: gradients of scalar fields are conservative vector fields. More generally, in a simply connected region,

$$\nabla \times \mathbf{F} = \mathbf{0} \quad \iff \quad \mathbf{F} = \nabla\phi \tag{1.1.69}$$

for some scalar potential ϕ, subject to the appropriate regularity conditions.

1.1.6.2 Surface Integrals

A surface integral is the analogous construction for a two-dimensional surface embedded in three-dimensional space. Let a surface Σ be parameterized by two parameters u and v,

$$\mathbf{r} = \mathbf{r}(u, v), \qquad (u, v) \in D. \tag{1.1.70}$$

The tangent vectors to the surface are

$$\frac{\partial \mathbf{r}}{\partial u}, \qquad \frac{\partial \mathbf{r}}{\partial v}, \tag{1.1.71}$$

and therefore the infinitesimal area element is

$$dS = \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du\, dv. \tag{1.1.72}$$

For a scalar field ϕ, the surface integral is

Definition 1.12 : Scalar surface integral

$$\int_{\Sigma} \phi \; dS = \iint_{D} \phi(\mathbf{r}(u, v)) \left\| \frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right\| du\, dv. \tag{1.1.73}$$

This is the two-dimensional analogue of the scalar line integral: rather than summing the field along a curve, we sum it over an entire surface, weighted by the infinitesimal area element.

For a vector field $\mathbf{F}$, however, the natural surface integral is a *flux integral.* The oriented surface element is defined by

$$d\mathbf{S} = \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du\, dv, \tag{1.1.74}$$

where the direction of $d\mathbf{S}$ is determined by the chosen orientation of the surface. The flux of $\mathbf{F}$ through Σ is then

Definition 1.13 : Vector surface integral

$$\int_\Sigma \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u,v)) \cdot \left(\frac{\partial \mathbf{r}}{\partial u} \times \frac{\partial \mathbf{r}}{\partial v} \right) du\, dv\,. \tag{1.1.75}$$

The dot product selects the component of the vector field normal to the surface. Thus, the flux measures how much of the field passes through the surface rather than how much of it runs along the surface.

1.1.6.3 Volume Integrals

Finally, a volume integral is an integral over a three-dimensional region V. For a scalar field ϕ, we write

$$\int_V \phi\, dV. \tag{1.1.76}$$

In Cartesian coordinates,

$$dV = dx\, dy\, dz, \tag{1.1.77}$$

so that

Definition 1.14 : Volume integral

$$\int_V \phi\, dV = \iiint_V \phi(x,y,z)\, dx\, dy\, dz. \tag{1.1.78}$$

These constructions form the basis for the integral versions of the fundamental theorems of vector calculus, including Green's theorem, Stokes' theorem, and the divergence theorem. These theorems relate derivatives of a field within some region to the behavior of that field on the boundary of the region.

1.1.7 Gauss and Stokes Theorem

Gauss's and Stokes' theorems are two of the most important results in vector calculus, providing relationships between surface and volume (or line and surface) integrals. They serve as generalizations of the Fundamental Theorem of Calculus to higher dimensions, allowing one to convert integrals over regions into integrals over their boundaries.

1.1.7.1 Gauss's Divergence Theorem

The divergence theorem, also known as Gauss's theorem, relates the flux of a vector field through a closed surface to the divergence of the field inside the volume enclosed by that surface. Mathematically, it is expressed as

Theorem 1.2 : Gauss' Theorem

$$\iiint_V (\nabla \cdot \mathbf{F})\, dV = \oiint_S \mathbf{F} \cdot d\mathbf{S}, \tag{1.1.79}$$

where V is the volume enclosed by the closed surface S, $\mathbf{F}$ is a continuously differentiable vector field, and $d\mathbf{S}$ is the outward-pointing area element on S, sometimes denoted $\hat{\mathbf{n}}\, dS$, where $\hat{\mathbf{n}}$ is the surface normal unit vector on the surface. This theorem is widely used in electromagnetism, fluid mechanics, and gravitational theory.

The divergence $\nabla \cdot \mathbf{F}$ measures the local tendency of the vector field to act as a source or sink. A positive divergence indicates that, locally, more field is flowing out of a small volume than into it, while a negative divergence indicates a net inward flow. The divergence theorem states that the total of these local sources and sinks throughout a volume is equal to the net outward flux through its boundary. Thus, it provides a direct connection between a local differential quantity and a global integral quantity. The theorem requires the surface to be closed, since every point on the boundary must enclose a volume over which the divergence can be integrated.

This is sometimes shortened as

$$\int_\Omega \nabla \cdot \mathbf{F}\, dV = \oint_{\partial\Omega} \mathbf{F} \cdot d\mathbf{S} \tag{1.1.80}$$

1.1.7.2 Stokes' Theorem

Stokes' theorem relates the circulation of a vector field around a closed curve to the curl of the field over the surface bounded by that curve. It can be written as

Theorem 1.3 : Stokes' Theorem

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot d\mathbf{S}, \tag{1.1.81}$$

where C is the positively oriented boundary curve of the surface S. Stokes' theorem generalizes the concept of Green's theorem in the plane to three dimensions.

The quantity $\nabla \times \mathbf{F}$, called the curl of the vector field, measures the local tendency of the field to produce circulation about an axis. The line integral on the left measures the total circulation of the field around the boundary curve C. Stokes' theorem therefore relates the local rotational behavior of a vector field throughout a surface to the total circulation observed along its boundary. Unlike the divergence theorem, which concerns flux through a closed surface, Stokes' theorem concerns circulation around a closed curve and the surface spanning that curve.

Remark :

In essence, both theorems express how local differential properties of vector fields (divergence and curl) are related to global integral quantities (flux and circulation). They form the mathematical foundation for many physical laws, such as Gauss's law in electromagnetism and the conservation of mass and momentum in fluid dynamics.

1.1.8 Dirac Delta Point Source

The **Dirac delta function** $\delta(\mathbf{r})$ (or $\delta^{(3)}(\mathbf{r})$ for three dimensions) models an idealized point source.

Definition 1.15 : Dirac Delta function

$$\delta(x) = \begin{cases} 1 & x = 0 \\ 0 & x \neq 0 \end{cases} \tag{1.1.82}$$

It has the property

$$\int_{-\infty}^{+\infty} \delta(x)\, dx = 1 \tag{1.1.83}$$

Its defining property is the sifting integral:

Definition 1.16 : Dirac Delta Sifting

$$\int_V f(\mathbf{r})\, \delta(\mathbf{r} - \mathbf{r}_0)\, d^3r = f(\mathbf{r}_0), \tag{1.1.84}$$

for any test function f and any volume V containing $\mathbf{r}_0$. The delta is zero for $\mathbf{r} \neq \mathbf{r}_0$, singular at $\mathbf{r} = \mathbf{r}_0$, and normalized so that its integral over all space is 1:

$$\int_{\mathbb{R}^3} \delta(\mathbf{r})\, d^3r = 1. \tag{1.1.85}$$

Delta and differential operators The Dirac delta often appears as the source term in Poisson-like equations. Two classic identities (in three dimensions) are:

$$\nabla^2 \left(\frac{1}{r} \right) = -4\pi\, \delta^{(3)}(\mathbf{r}), \tag{1.1.86}$$

$$\nabla \cdot \left(\frac{\mathbf{r}}{r^3} \right) = 4\pi\, \delta^{(3)}(\mathbf{r}). \tag{1.1.87}$$

These relations are to be understood in the distributional sense: the left-hand side vanishes for $\mathbf{r} \neq \mathbf{0}$ but the integral over any volume containing the origin produces the stated constant.

1.1.9 Potential fields

1.1.9.1 Irrotational fields

If the curl of a vector field **F** vanishes everywhere, then **F** can be written as the gradient of a scalar potential field V.

$$\nabla \times \mathbf{F} = \mathbf{0} \iff \mathbf{F} = -\nabla V \tag{1.1.88}$$

Theorem 1.4 : Curl-less or irrotational fields

The following conditions are equivalent.

(a) $\nabla \times \mathbf{F} = \mathbf{0}$

(b) $\displaystyle\int_a^b \mathbf{F} \cdot d\boldsymbol{\ell}$ is independent of path for any given end points.

(c) $\displaystyle\oint \mathbf{F} \cdot d\boldsymbol{\ell} = 0$

(d) $\mathbf{F} = -\nabla V$

1.1.9.2 Solenoidal Fields

If the divergence of a vector field **F** vanishes everywhere, then **F** can be written as the curl of a vector potential field **A**.

$$\nabla \cdot \mathbf{F} = 0 \iff \mathbf{F} = \nabla \times \mathbf{A} \tag{1.1.89}$$

Theorem 1.5 : Divergence-less or solenoidal fields

The follorwing conditions are equivalent.

(a) $\nabla \cdot \mathbf{F} = 0$

(b) $\displaystyle\int \mathbf{F} \cdot d\mathbf{s}$ is independent of surface for any given boundary line.

(c) $\displaystyle\oint \mathbf{F} \cdot d\mathbf{s} = 0$

(d) $\mathbf{F} = \nabla \times \mathbf{A}$

Remark :

A vector field potential is not restricted to these definitions. A vector field $\mathbf{F}$ can be written as the sum of a gradient of a scalar and a curl of a vector field.

$$\mathbf{F} = -\nabla V + \nabla \times \mathbf{A} \tag{1.1.90}$$

1.1.10 Curvilinear coordinates

Cartesian coordinates are not the only way to describe position in space. More generally, we may specify a point using three coordinates (q_1, q_2, q_3) whose coordinate surfaces need not be planes. Such a system is called a *curvilinear coordinate system.* The familiar cylindrical and spherical coordinates are important examples.

We introduce an orthogonal curvilinear coordinate system (q_1, q_2, q_3) with mutually perpendicular unit vectors $\hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2, \hat{\mathbf{e}}_e$ and scale factors (metric coefficients)

$$h_i(q_1, q_2, q_3) = \left| \frac{\partial \mathbf{r}}{\partial q_i} \right|, \qquad i = 1, 2, 3.$$

The scale factor h_i measures the physical distance corresponding to a small change dq_i in the coordinate q_i. Consequently, the infinitesimal displacement is

$$d\mathbf{r} = h_1\, dq_1\, \hat{\mathbf{e}}_1 + h_2\, dq_2\, \hat{\mathbf{e}}_2 + h_3\, dq_3\, \hat{\mathbf{e}}_3$$

The important difference from Cartesian coordinates is that the basis vectors $\hat{\mathbf{e}}_i$ and the scale factors h_i can depend on position. In Cartesian coordinates,

$$h_x = h_y = h_z = 1,$$

and the basis vectors are constant throughout space. In curvilinear coordinates, however, the direction of the basis vectors can change as we move from one point to another.

For an orthogonal coordinate system, a small coordinate volume dq_1, dq_2, dq_3 corresponds to the physical volume

$$dV = h_1 h_2 h_3\, dq_1\, dq_2\, dq_3\,.$$

Similarly, the area element of a surface on which q_i is constant is

$$dA_i = \frac{h_1 h_2 h_3}{h_i} dq_j dq_k, \qquad i \neq j \neq k.$$

Thus, the scale factors determine not only distances but also the geometry of infinitesimal areas and volumes.

Gradient. The gradient of a scalar field ϕ is

Definition 1.17 : Gradient in General Curvilinear Coordinates

$$\nabla\phi = \sum_i \frac{1}{h_i} \frac{\partial \phi}{\partial q_i} \hat{\mathbf{e}}_i \tag{1.1.91}$$

The factors $1/h_i$ are required because $\partial\phi/\partial q_i$ describes the change in ϕ per unit change in the coordinate q_i, whereas the gradient describes change per unit *physical distance.* Since

$$ds_i = h_i \, dq_i \,,$$

we have

$$\frac{\partial \phi}{\partial s_i} = \frac{1}{h_i} \frac{\partial \phi}{\partial q_i}.$$

Divergence. The divergence measures the net outward flux of a vector field per unit volume. In orthogonal curvilinear coordinates,

Definition 1.18 : Divergence in general curvilinear coordinates

$$\nabla \cdot \mathbf{F} = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^{3} \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i} F_i \right). \tag{1.1.92}$$

The factors involving h_i account for the fact that the areas and volumes associated with the coordinate directions may change from point to point. In Cartesian coordinates, where all $h_i = 1$, this reduces immediately to the familiar definition of the divergence.

Curl. The curl measures the local tendency of a vector field to circulate around a point. For an orthogonal curvilinear coordinate system,

Definition 1.19 : Curl in general curvilinear coordinates

$$\nabla \times \mathbf{F} = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{\mathbf{e}}_1 & h_2 \hat{\mathbf{e}}_2 & h_3 \hat{\mathbf{e}}_3 \\ \dfrac{\partial}{\partial q_1} & \dfrac{\partial}{\partial q_2} & \dfrac{\partial}{\partial q_3} \\ h_1 F_1 & h_2 F_2 & h_3 F_3 \end{vmatrix} \tag{1.1.93}$$

Equivalently, its components are

$$(\nabla \times \mathbf{F})_1 = \frac{1}{h_2 h_3} \left[\frac{\partial}{\partial q_2}(h_3 F_3) - \frac{\partial}{\partial q_3}(h_2 F_2) \right] \tag{1.1.94}$$

$$(\nabla \times \mathbf{F})_2 = \frac{1}{h_3 h_1} \left[\frac{\partial}{\partial q_3}(h_1 F_1) - \frac{\partial}{\partial q_1}(h_3 F_3) \right] \tag{1.1.95}$$

$$(\nabla \times \mathbf{F})_3 = \frac{1}{h_1 h_2} \left[\frac{\partial}{\partial q_1}(h_2 F_2) - \frac{\partial}{\partial q_2}(h_1 F_1) \right]. \tag{1.1.96}$$

Laplacian. The Laplacian of a scalar field is defined as the divergence of its gradient:

$$\nabla^2 \phi = \nabla \cdot (\nabla \phi).$$

Using the gradient and divergence formulas above gives

Definition 1.20 : Laplacian in general curvilinear coordinates

$$\nabla^2 \phi = \frac{1}{h_1 h_2 h_3} \sum_{i=1}^{3} \frac{\partial}{\partial q_i} \left(\frac{h_1 h_2 h_3}{h_i^2} \frac{\partial \phi}{\partial q_i} \right). \tag{1.1.97}$$

This is the standard form of the scalar Laplacian in an orthogonal curvilinear coordinate system.

Using all these definitions of vector calculus in curvilinear (or general) coordinates, one would be able to derive the relevant values for other coordinate systems, provided that the scale factors for each coordinate system is known. Later on, we will realize that this all has something to do with what is called the *metric*, but that is for a General Relativity class, and this is not a general relativity class.

1.1.10.1 Cylindrical coordinates

Cylindrical coordinates (r, θ, z) describe a point by its distance r from the z-axis, its azimuthal angle θ about the z-axis, and its height z. They are related to Cartesian coordinates by

$$x = r\cos\theta, \qquad y = r\sin\theta, \qquad z = z,$$

or equivalently

$$\mathbf{r} = (r\cos\theta, r\sin\theta, z).$$

The corresponding local unit vectors are

$$\begin{aligned}
\hat{\mathbf{e}}_r &= (\cos\theta, \sin\theta, 0)\\
\hat{\mathbf{e}}_\theta &= (-\sin\theta, \cos\theta, 0)\\
\hat{\mathbf{e}}_z &= (0, 0, 1)
\end{aligned}$$

Unlike Cartesian basis vectors, $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ depend on θ and therefore change direction from point to point.

The scale factors are

$$h_r = 1 \qquad\qquad h_\theta = r \qquad\qquad h_z = 1$$

Thus

$$d\mathbf{r} = r\,dr\,\hat{\mathbf{e}}_r + r\,d\theta\,\hat{\mathbf{e}}_\theta + dz\,\hat{\mathbf{e}}_z$$

and

$$dV = r\,dr\,d\theta\,dz$$

A vector field is written as

$$\mathbf{F} = F_r\hat{\mathbf{e}}_r + F_\theta\hat{\mathbf{e}}_\theta + F_z\hat{\mathbf{e}}_z$$

Using (1.1.92), its divergence is

$$\nabla\cdot\mathbf{F} = \frac{1}{r}\frac{\partial}{\partial r}(rF_r) + \frac{1}{r}\frac{\partial F_\theta}{\partial\theta} + \frac{\partial F_z}{\partial z} \tag{1.1.98}$$

Using (1.1.93), its curl is

$$\nabla\times\mathbf{F} = \begin{bmatrix} \frac{1}{r}\frac{\partial F_z}{\partial\theta} - \dfrac{\partial F_\theta}{\partial z} \\ \dfrac{\partial F_r}{\partial z} - \dfrac{\partial F_z}{\partial r} \\ \dfrac{1}{r}\left(\dfrac{\partial}{\partial r}(rF_\theta) - \dfrac{\partial F_r}{\partial\theta}\right) \end{bmatrix} \tag{1.1.99}$$

Bear in mind that the components in vector component form are still relative to $\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta$, and $\hat{\mathbf{e}}_\phi$.

For the scalar Laplacian, $h_1 h_2 h_3 = r$, so

$$\begin{aligned}\nabla^2 \phi &= \frac{1}{r}\left[\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{\partial}{\partial \theta}\left(\frac{1}{r}\frac{\partial \phi}{\partial \theta}\right) + \frac{\partial}{\partial z}\left(r\frac{\partial \phi}{\partial z}\right)\right] \\ &= \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial \phi}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 \phi}{\partial \theta^2} + \frac{\partial^2 \phi}{\partial z^2}\end{aligned}$$

1.1.10.2 Spherical coordinates

Spherical coordinates (r, θ, φ) describe a point by its radial distance r from the origin, its polar angle θ measured from the positive z-axis, and its azimuthal angle φ measured around the z-axis.

They are related to Cartesian coordinates by

$$x = r\sin\theta\cos\varphi, \qquad y = r\sin\theta\sin\varphi, \qquad z = r\cos\theta,$$

so that

$$\mathbf{r} = \big(r\sin\theta\cos\varphi, , r\sin\theta\sin\varphi, , r\cos\theta\big).$$

The corresponding unit vectors are

$$\begin{aligned}\hat{\mathbf{e}}_r &= (\sin\theta\cos\varphi, \sin\theta\sin\varphi, \cos\theta) \\ \hat{\mathbf{e}}_\theta &= (\cos\theta\cos\varphi, \cos\theta\sin\varphi, -\sin\theta) \\ \hat{\mathbf{e}}_\varphi &= (-\sin\varphi, \cos\varphi, 0)\end{aligned}$$

Again, the basis is position-dependent. In particular, $\hat{\mathbf{e}}_r$ changes whenever either θ or φ changes, while $\hat{\mathbf{e}}_\varphi$ changes as φ changes.

The scale factors are

$$h_r = 1, \qquad h_\theta = r, \qquad h_\varphi = r\sin\theta.$$

Thus

$$d\mathbf{r} = dr\,\mathbf{e}_r + r\,d\theta\,\hat{\mathbf{e}}_\theta + r\sin\theta\,d\varphi\,\hat{\mathbf{e}}_\varphi$$

and

$$dV = r^2\sin\theta, dr\,, d\theta\,, d\varphi$$

A vector field is written as

$$\mathbf{F} = F_r\hat{\mathbf{e}}_r + F_\theta\hat{\mathbf{e}}_\theta + F_\varphi\hat{\mathbf{e}}_\varphi$$

Using (1.1.92), its divergence becomes

$$\nabla \cdot \mathbf{F} = \frac{1}{r^2}\frac{\partial}{\partial r}(r^2 F_r) + \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta\, F_\theta) + \frac{1}{r\sin\theta}\frac{\partial F_\varphi}{\partial\varphi} \tag{1.1.100}$$

Its curl is

$$\nabla \times \mathbf{F} = \begin{bmatrix} \frac{1}{r\sin\theta}\left(\frac{\partial}{\partial\theta}(\sin\theta F_\varphi) - \frac{\partial F_\theta}{\partial\varphi}\right) \\ \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial F_r}{\partial\varphi} - \frac{\partial}{\partial r}(rF_\varphi)\right) \\ \frac{1}{r}\left(\frac{\partial}{\partial r}(rF_\theta) - \frac{\partial F_r}{\partial\theta}\right) \end{bmatrix} \tag{1.1.101}$$

For the scalar Laplacian, $h_1h_2h_3 = r^2\sin\theta$. Substituting into (1.1.97) gives

$$\begin{aligned} \nabla^2\phi &= \frac{1}{r^2\sin\theta}\left[\frac{\partial}{\partial r}\left(r^2\sin\theta\frac{\partial\phi}{\partial r}\right) + \frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{\partial}{\partial\varphi}\left(\frac{1}{\sin\theta}\frac{\partial\phi}{\partial\varphi}\right)\right] \\ &= \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial\phi}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\phi}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2\phi}{\partial\varphi^2} \end{aligned}$$

1.1.10.3 Miscellaneous Geometries

There are a host of other geometries which may prove relevant in the study of vector calculus. To name a few, there are the parabolic coordinates (σ, τ, ϕ), Elliptic cylindrical (μ, ν, z), parabolic cylindrical (μ, ν, z), spheroidal (for oblate/prolate spheroids), (ξ, η, ϕ), and toroidal coordinate systems (η, ξ, ϕ). However, for the most part, cylindrical and spherical systems are the most prevalent systems one will encounter while working with majority of physics, so it is handy to have certain formulae in hand to get these quantities.

If the need arises to tackle these other bizarre geometries (if one were studying General Relativity, for example), then the defined gradient, curl, divergence, and laplacian formulae at the beginning of this section will be utilized, provided that the metric for the coordinate system is known. Then, deriving a specific formula for such coordinate system is trivial.

1.1.11 Divergence and the Dirac Delta

As is required from the course notes goals, we aim to prove that the divergence of a radial inverse square vector field is consistent with the Dirac delta point source/sink.

Consider the radial inverse-square vector field

$$\mathbf{F}(\mathbf{r}) = \alpha \frac{1}{r^2}\hat{\mathbf{r}} = \alpha \frac{\mathbf{r}}{r^3} \tag{1.1.102}$$

where $r = \|\mathbf{r}\|$, $\hat{\mathbf{r}} = \mathbf{r}/r$, and $\alpha \in \mathbb{R}$ is constant.

We want to show that the divergence is zero on this field everywhere except at the origin, where it is represented by a Dirac delta:

$$\nabla \cdot \mathbf{F} = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \tag{1.1.103}$$

For $\alpha > 0$, this is a point source, and for $\alpha < 0$, a point sink.

Proof. We prove the relation with the following arguments:

1. **Divergence away from the origin**
   For a purely radial field

   $$\mathbf{F} = F_r(r)\hat{\mathbf{r}} \tag{1.1.104}$$

   the divergence in spherical coordinates is

   $$\nabla \cdot \mathbf{F} = \frac{1}{r^2}\frac{d}{dr}\left(r^2 F_r\right) \tag{1.1.105}$$

   Here, the magnitude of F_r is

   $$F_r = \frac{\alpha}{r^2} \tag{1.1.106}$$

   so,

   $$r^2 F_r = \alpha \tag{1.1.107}$$

   and therefore, for $r \neq 0$,

   $$\begin{aligned} \nabla \cdot \mathbf{F} &= \frac{1}{r^2}\frac{d\alpha}{dr} && (1.1.108)\\ &= \frac{1}{r^2} \cdot 0 && (1.1.109)\\ &= 0. && (1.1.110) \end{aligned}$$

Thus,

$$\nabla \cdot \mathbf{F} = 0 \quad (r \neq 0). \tag{1.1.111}$$

Therefore, any nonzero divergence must be concentrated at $r = 0$.

2. **Flux through a sphere about the origin**

   Suppose a sphere S_R of radius R centered at the origin. Since

   $$\mathbf{F} = \frac{\alpha}{R^2}\hat{\mathbf{r}} \tag{1.1.112}$$

   on the sphere, and

   $$d\mathbf{A} = \hat{\mathbf{r}} R^2 \sin\theta \, d\theta \, d\phi \,, \tag{1.1.113}$$

   the flux is

   $$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = \int_0^{2\pi} \int_0^{\pi} \frac{\alpha}{R^2} R^2 \sin\theta \, d\theta \, d\phi \,, \tag{1.1.114}$$

   Evaluating this, we have

   $$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = \alpha \cdot (4\pi) \tag{1.1.115}$$

   Therefore,

   $$\oint_{S_R} \mathbf{F} \cdot d\mathbf{A} = 4\pi\alpha \tag{1.1.116}$$

   Note that this is independent of R.

3. **Gauss' Law**

   For an ordinary smooth field, the divergence theorem gives

   $$\int_V \nabla \cdot \mathbf{F} \, d^3r = \oint_{\partial V} \mathbf{F} \cdot d\mathbf{A} \tag{1.1.117}$$

   Thus, for any volume containig the origin,

   $$\int_V \nabla \cdot \mathbf{F} \, d^3r = 4\pi\alpha \tag{1.1.118}$$

   From argument 1, we already established that the divergence is zero everywhere except the origin. Thus, the entire $4\pi\alpha$ must be concetrated at the single point $\mathbf{r} = 0$.

As such, we desire a function $g(\mathbf{r})$ that satisfies

$$g(\mathbf{r}) = 0, \quad \mathbf{r} \neq 0 \tag{1.1.119}$$

but

$$\int_V g(\mathbf{r}) d^3r = 4\pi\alpha \tag{1.1.120}$$

Whenever V containts the origin.

4. **Property of the Dirac Delta**

   The three dimensional Dirac delta is defined by

$$\int_V \delta^{(3)}(\mathbf{r}) d^3r = \begin{cases} 1, & 0 \in V \\ 0, & 0 \notin V \end{cases} \tag{1.1.121}$$

   Therefore, $4\pi\alpha\delta^{(3)}(\mathbf{r})$ has the property

$$\int_V 4\pi\alpha\delta^{(3)}(\mathbf{r}) d^3r = \begin{cases} 4\pi\alpha & 0 \in V \\ 0 & 0 \notin V \end{cases} \tag{1.1.122}$$

   Which is precisely what we require of our function $g(\mathbf{r})$

   Therefore, in the sense of distributions,

$$\nabla \cdot \mathbf{F} = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \tag{1.1.123}$$

Since $\mathbf{F} = \dfrac{\alpha}{r^2}\hat{\mathbf{r}}$, then we find that

$$\nabla \cdot \left(\alpha \frac{1}{r^2}\hat{\mathbf{r}}\right) = 4\pi\alpha\delta^{(3)}(\mathbf{r}) \tag{1.1.124}$$

We can cancel the α, since it's simply a constant indicating source/sink dynamics. Doing this, we find that

$$\nabla \cdot \left(\frac{1}{r^2}\hat{\mathbf{r}}\right) = 4\pi\delta^{(3)}(\mathbf{r}) \tag{1.1.125}$$

Which is what we set out to prove. $\square$

Thus, from these results, we have the following theorems:

Theorem 1.6 : Divergence of a radial inverse quadratic vector field

The divergence of a radial inverse quadratic vector field is consistent with the Dirac delta point source/sink

$$\nabla \cdot \left(\frac{1}{r^2}\hat{\mathbf{r}}\right) = 4\pi\delta^{(3)}(\mathbf{r}) \tag{1.1.126}$$

A particular consequence of this is that the Laplacian of the potential field $\phi = \frac{1}{r}$ (whose gradient field is $\mathbf{F} = \frac{1}{r^2}\hat{\mathbf{r}}$) is also a Dirac delta.

$$\begin{aligned}
\nabla\left(\frac{1}{r}\right) &= -\frac{1}{r^2}\hat{\mathbf{r}} \\
\nabla \cdot \nabla\left(\frac{1}{r}\right) &= -\nabla \cdot \left(\frac{1}{r^2}\hat{\mathbf{r}}\right) \\
\nabla^2\left(\frac{1}{r}\right) &= -4\pi\delta^{(3)}(\mathbf{r})
\end{aligned}$$

Theorem 1.7 : Laplacian of an radial inverse potential field

$$\nabla^2\left(\frac{1}{r}\right) = -4\pi\delta^{(3)}(\mathbf{r}) \tag{1.1.127}$$

1.2 The Electric Field

1.2.1 Definition

The fundamental problem that electrodynamic hopes to solve is this: We have some collection of electric charges $q_1, q_2, q_3, \ldots$, called *source charges*; What force do they exert on another charge q_0, called *text charge*? The positions of the source charges are given (all as functions of time) and the trajectory of the test particle is *to be calculated.* In general, both the source charges and test charge are in motion.

The solution to this problem is facilitated by the **principle of superposition**, which states that the interaction between any two charges is completely unnafected by the presence of others. That means that to determine the force on q_0, we must first determine the force $\mathbf{F}_1$ due to q_1 alone, then q_2, and so on. Finally, we take the vector sum of these individual forces.

$$\mathbf{F} = \sum_i \mathbf{F}_i = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \cdots$$

Thus, if we can find the force on q_0 due to a *single* source charge q, we are, in principle, done, and the rest is just repitition.

A priori, this seems easy. However, upon further study, we find that the force on q_0 not only depends on the separation distance r between the charges, but also depends on both their velocities and on the acceleration of q. Moreover, it is not those states at this instant that matters, but the state that q had in an earlier time, since information travels at a finite speed c, and we can only grasp what happened after the information reached us.

Therefore, in spite of the fact that the basic question ("What is the force on q_0 due to q?") is easy to state, it does not pay to confront it head on; rather, we shall go at it by stages. In the meantime, the theory we develop will allow for the solution of more subtle electromagnetic problems that do not present themselves in quite this simple format. To begin with, we shall consider the special case of **electrostatics** in which all the *source charges are stationary* (though the test charge may be moving).

1.2.2 Coulomb's Law

The force on a test charge q_0 due to a single point charge q, that is at *rest* a distance r away, is given (based on experiments) by **Coulomb's Law**, a fundamental principle first established by Charles-Augustin de Coulomb in 1785 using a torsion balance experiment.

It is expressed as the following:

Definition 1.21 : Coulomb's Law

$$\mathbf{F} = \frac{1}{4\pi\varepsilon_0}\frac{q_0 q}{r^2}\hat{\mathbf{r}} \tag{1.2.1}$$

The constant ε_0 is called the **permitivity of free space**. In SI units, it is defined as

Definition 1.22 : Permitivity of free space

$$\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{C}^2/\mathrm{Nm}^2 \tag{1.2.2}$$

In other words, then, the force is proportional to the product of the charges, and inversely proportional to the square of their distances[2]. The vector $\mathbf{r}$ is the separation vector from $\mathbf{x}$, the location of q, and to $\mathbf{x}_0$, the location of the q_0

$$\mathbf{r} = \mathbf{x} - \mathbf{x}_0$$

and r is the magnitude of this separation vector, $r = \|\mathbf{r}\|$. The force points along the line from q to q_0; it is repulsive if q_0 and q are the same sign, and attractive if their signs are the opposite.

Coulomb's law and the principle of superposition constitute the physical input for electrostatics—the rest, except for some special properties of matter, is mathematical elaboration of these fundamental rules.

[2] Newton much?

1.2.3 The Electric Field

If we have *several* point charges $q_1, q_2, q_3, \ldots, q_n$, at distances $r_1, r_2, r_3, \ldots, r_n$ from q_0, then the total force on q_0 is trivially

$$\begin{aligned}
\mathbf{F} &= \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \cdots \\
&= \frac{1}{4\pi\varepsilon_0}\left(\frac{q_0 q_1}{r_1^2}\hat{\mathbf{r}}_1 + \frac{q_0 q_2}{r_2^2}\hat{\mathbf{r}}_2 + \frac{q_0 q_3}{r_3^2}\hat{\mathbf{r}}_3 + \cdots\right) \\
&= \frac{q_0}{4\pi\varepsilon_0}\left(\frac{q_1}{r_1^2}\hat{\mathbf{r}}_1 + \frac{q_2}{r_2^2}\hat{\mathbf{r}}_2 + \frac{q_3}{r_3^2}\hat{\mathbf{r}}_3 + \cdots\right)
\end{aligned}$$

or

$$\mathbf{F} = q_0 \mathbf{E} \tag{1.2.3}$$

where

Definition 1.23 : Discrete definition of the Electric field

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\sum_{i=1}^{n}\frac{q_i}{r_i^2}\hat{\mathbf{r}}_i \tag{1.2.4}$$

E is called the **electric field** of the source charges. Notice that it is a function of position ($\mathbf{r}$) because the separation vectors $\mathbf{r}$ depend on the location of the field point P. But it makes no reference to the test charge q. The electric field is a vector quantity which varies from point to point and is determined by the configuration of source charges. Physically ,$\mathbf{E}(\mathbf{r})$ is the force per unit charge that would be exerted on a text charge if you were to place one on P.

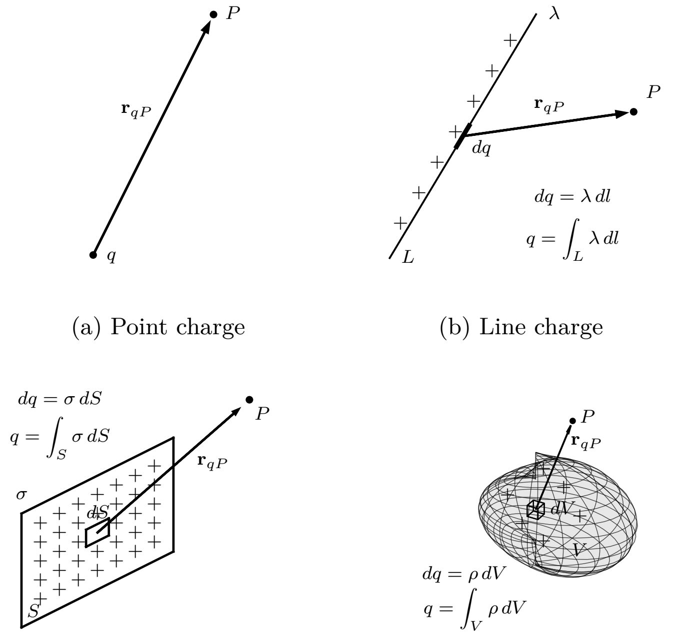
The principle of superposition also means that the electric field produced by a collection of charges can be determined independently of any other charges that may subsequently be introduced into the system. Each source charge contributes its own electric field at the point P, and the total field is simply the vector sum of these individual contributions. This is particularly useful because the field can be calculated once for a given configuration of source charges and then used to determine the force on any test charge placed within that field.

1.2.4 Continuus Charge Distribution

Our definition of the electric field (1.2.4) assumes that the source of the fields is a collection of discrete point charges q_i (Fig 1.6a). If instead, the charge is distributed continuously over some region, the sum thus becomes an integral

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{1}{r^2} \hat{\mathbf{r}} \, dq \tag{1.2.5}$$

There are multiple ways where in charges can be distributed over a region. In essence, for every dimension (0,1,2,3), there is a corresponding region where charges can be distributed, and where charges can be effectively calculated.



(a) Point charge

(b) Line charge

(c) Surface charge

(d) Volume charge

Figure 1.6: Charge distributions

If the charge is spread out along a *line* (Fig 1.6b), with the charge-per-unit-length λ, then $dq = \lambda\, dl$, where dl is a length element along the line. If the charge is smeared over a *surface* (Fig 1.6c), with charge-per-unit-area σ, then $dq = \sigma\, dS$, where dS is an area element on the surface. And if the charge fills a *volume* (Fig 1.6d), with charge-per-unit-volume ρ, then $dq = \rho\, dV$, where dV is a volume element.

$$dq \to \lambda dl \sim \sigma dS \sim \rho dV$$

Thus, the electric field of a line charge is

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda(\mathbf{r})}{r^2} \hat{\mathbf{r}}\, dl\,; \tag{1.2.6}$$

For a surface charge, we have

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\sigma(\mathbf{r})}{r^2} \hat{\mathbf{r}}\, dS\,; \tag{1.2.7}$$

And for a volume charge

Definition 1.24 : Continuum definition of the Electric Field

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r})}{r^2} \hat{\mathbf{r}}\, dV\,; \tag{1.2.8}$$

Equation 1.2.8 is often times referred to as "Coulomb's Law" because it is such a short step from the original, and because a volume charge is in a sense the most general and realisitic case.

Note carefully the meaning of $\mathbf{r}$ in these formulae. In general, $\mathbf{r}$ pertains to the vector from the source charge q to the field point P, which means, the meain distance from $dl\,, dS\,, dV$ from the point P. Although, since these regions are differentials, they reduce to almost pointlike at the limits, so the analogy of *distance from the source point to the field point* still stands.

1.2.5 Divergence and Curl of Electrostatic Fields

We begin with the simplest possible case of electrostatics. A single point q, situataed at the origin.

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$

We then have a plot which looks akin to this, the field lines of this point source[3] These field lines are indicative of the strength of the field.

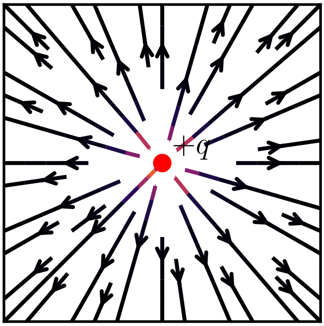


Figure 1.7: Electric field lines

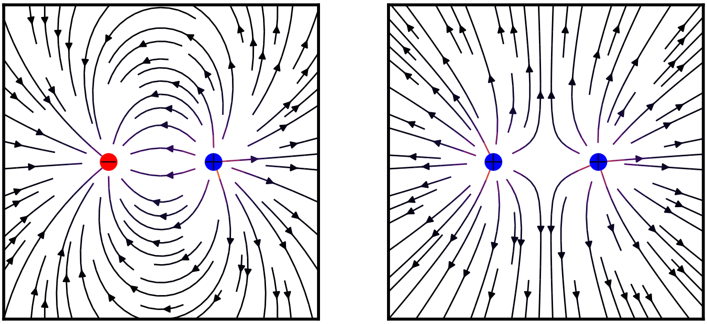
More field lines mean a stronger field, denser field lines imply larger field strength, and so on. Such diagrams are convinient for representing more complicated fields. Field lines begin from positive charges and end on negative charges, and cannot simply terminate mid-field, though, they might extend from infinity. Moreover, field lines can never cross, for at such an intersection, the field would have two different directions at once. With all these in mind, it is easy to sketch out the field of any simple configuration of point charges. Begin by drawing the line from the positive, calling it a *source*, and ending it on the negative point, the *sink*.

In this model, the *flux* of $\mathbf{E}$ through a surface S

$$\Phi_E = \int_S \mathbf{E}\cdot d\mathbf{S}$$

[3] Unfortunately `matplotlib` does not have a convenient field line functionality so we have to do `streamplot` for now

is a measure of the number of field lines passing through S[4]. Though in this notes, we will mainly be showing stream plots, which show the flow of the vector field, instead of actual field lines, due to aforementioned limitations. However, in general, the strength of the field is proportional to the number of lines through a given sampling area. The dot product $\mathbf{E} \cdot d\mathbf{S}$ picks out the components of $d\mathbf{S}$ which is along the direction of $\mathbf{E}$



(a) Opposite charges (b) Equal charges

Figure 1.8: Charge interactions

This suggests that the flux through any closed surface is a measure of the total charge inside. For the field lines that originate on a positive charge must either pass out through the surface or else terminate on a negative charge inside (Fig 1.8a). On the other hand, a charge outside the surface will contribute nothing to the total flux, since its field lines pass in one side and out the other (Fig 1.8b). This is the *essence* of Gauss's law.

Quantitatively, this can be expressed as the following: Suppose a point charge q at the origin. The flux of $\mathbf{E}$ through a spherical surface of radius r is:

$$\oint \mathbf{E} \cdot d\mathbf{S} = \iint \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r^2}\hat{\mathbf{r}}\right) \cdot \left(r^2 \sin\theta \, d\theta \, d\phi \, \hat{\mathbf{r}}\right) \tag{1.2.9}$$

$$= \frac{1}{4\pi\varepsilon_0} \cdot 4\pi q \tag{1.2.10}$$

$$= \frac{1}{\varepsilon_0} q \tag{1.2.11}$$

[4] $d\mathbf{S}$ in this case pertains to $\hat{\mathbf{n}}\, dS$

Notice that the radius of the sphere cancels out, for while the surface area goes up as r^2, the field strength goes down as $\frac{1}{r^2}$, and as such the product is constant, regardless of the size of the sphere. While we approach this from a spherical picture centered at the origin, it is notable that any closed surface, regardless of its shape, would be pierced by the same number of field lines. As such, the flux through any surface is constant, q/ε_0.

Now, suppose that instead of a single charge at the origin, we have a group of charges scattered around. According to the principle of superposition, the total field is the sum of all the individual fields.

$$\mathbf{E} = \sum_i \mathbf{E}_i$$

Thus, the flux through a surface that encloses them all is

$$\begin{aligned}\oint \mathbf{E} \cdot d\mathbf{S} &= \sum_i \left(\oint \mathbf{E}_i \cdot d\mathbf{S} \right) \\ &= \sum_i \left(\frac{1}{\varepsilon_0} q_i \right)\end{aligned}$$

For any closed surface, then

$$\oint \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0} Q_{\text{enc}} \tag{1.2.12}$$

where Q_{enc} is the total charge enclosed within the surface. This is the quantitative statement of Gauss' law.

As it stands, Gauss's law is an *integral* equation, but one can easily turn it into a *differential* one by applying the divergence theorem

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{E})\, dV \tag{1.2.13}$$

We rewrite Q_{enc} in terms of the charge density ρ, we have

$$Q_{\text{enc}} = \int_V \rho\, dV \tag{1.2.14}$$

So Gauss's law becomes

$$\int_V (\nabla \cdot \mathbf{E})\, dV = \int_V \left(\frac{\rho}{\varepsilon_0} \right) dV \tag{1.2.15}$$

Since this holds for any volume, this implies that the integrands are equal, hence:

Theorem 1.8 : Gauss's Law

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \tag{1.2.16}$$

This is Gauss's law in differential form. The differential version is more compact and tidier, but the integral form has the advantage in that it accomodates point, line, and surface charges more naturally.

1.2.5.1 The Divergence of E

Let us then return and calculate the divergence of $\mathbf{E}$ diretly from Eq. 1.2.8

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{r})}{r^2} \hat{\mathbf{r}}\, dV \tag{1.2.17}$$

Note that $\mathbf{r} = \mathbf{x} - \mathbf{q}$, where $\mathbf{x}$ is the point being evaluated and $\mathbf{q}$ is the source point. We note the $\mathbf{x}$ independence in $\mathbf{r}$. We apply the divergence operator on this field. Thus:

$$\nabla \cdot \mathbf{E} = \frac{1}{4\pi\varepsilon_0} \int \nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) \rho(\mathbf{r})\, dV \tag{1.2.18}$$

Using Eq. 1.1.126, we know that

$$\nabla \cdot \left(\frac{\hat{\mathbf{r}}}{r^2} \right) = 4\pi\delta^3(\mathbf{r})$$

Thus

$$\begin{aligned} \nabla \cdot \mathbf{E} &= \frac{1}{4\pi\varepsilon_0} \int 4\pi\delta^3(\mathbf{x} - \mathbf{q})\rho(\mathbf{q})\, dV \\ &= \frac{1}{\varepsilon_0}\rho(\mathbf{x}) \end{aligned}$$

This is simply Gauss's law in differential form. To recover the integral form, we run the previous steps in reverse, integrating over a volume and applying the divergence theorem

$$\int_V \nabla \cdot \mathbf{E}\, dV = \oint \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0} \int_V \rho\, dV = \frac{1}{\varepsilon_0} Q_{\text{enc}} \tag{1.2.19}$$

1.2.5.2 The Curl of E

We calculate the curl of **E** by studying first the simplest possible configuration: a point charge at the origin. In this case, we have

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\hat{\mathbf{r}}$$

A single glance at Fig. 1.7 should tell one that the curl of this field has to be zero. However, we aim to be rigourous, and so, suppose we calculate the line integral of this field from some point **a** to another point **b**.

$$\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\boldsymbol{\ell}$$

In spherical coordinates, $d\boldsymbol{\ell} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$, therefore

$$\mathbf{E}\cdot d\boldsymbol{\ell} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,dr$$

Therefore:

$$\begin{aligned}
\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\boldsymbol{\ell} &= \frac{1}{4\pi\varepsilon_0}\int_{\mathbf{a}}^{\mathbf{b}}\frac{q}{r^2}\,dr \\
&= -\frac{1}{4\pi\varepsilon_0}\left.\frac{q}{r}\right|_{r_a}^{r_b} \\
&= \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_a}-\frac{q}{r_b}\right)
\end{aligned}$$

where r_a is the distance of the origin from **a** and r_b is the distance of the origin from **b**. The integral around a *closed* path is evidently zero (because $r_a = r_b$), hence:

$$\oint \mathbf{E}\cdot d\boldsymbol{\ell} = 0 \tag{1.2.20}$$

from this, we apply Stokes's theorem

$$\nabla\times\mathbf{E} = \mathbf{0} \tag{1.2.21}$$

Proving this for a single point charge at the origin would suffice, but it all seems rather arbitrary. However, these results should hold no matter *where* the charge is located. Moreover, if there are multiple charges, the principle of superposition states that the total field is the vector sum of all the individual fields. Thus, we have

$$\begin{aligned}
\nabla \times \mathbf{E} &= \nabla \times \sum_i \mathbf{E}_i \\
&= \nabla \times (\mathbf{E}_1 + \mathbf{E}_2 + \cdots) \\
&= (\nabla \times \mathbf{E}_1) + (\nabla \times \mathbf{E}_2) + \cdots \\
&= \mathbf{0} + \mathbf{0} + \cdots \\
&= \mathbf{0}
\end{aligned}$$

Hence, we have proven that the curl of an electrostatic field, as a consequence of Coulomb's law, is zero everywhere. From this we have the electrostatic condition

Theorem 1.9 : Electrostatic condition

$$\nabla \times \mathbf{E} = \mathbf{0} \tag{1.2.22}$$

Remark :

We can also prove this by showing that the electric field arises from the gradient of a scalar field potential $\mathbf{E} = -\nabla\Phi$ by the inverse quadratic nature of a point source, and as such is conservative. From theorem 1.4, it is then trivial to show that $\nabla \times \mathbf{E} = \mathbf{0}$.

1.2.6 Dipoles

As another part of what we need to prove in the course notes, we have the following problem.

Consider a dipole with charge q with separation distance a and dipole moment $p = qa$. Evaluate the electric potential and electric field due to this configuration. Then take the point dipole limit $a \to 0$ with p finite. Find the far-field electric potential and electric field.

Solution Suppose that the dipole is oriented along the z-axis with $+q$ at $z = +a/2$ and $-q$ at $z = -a/2$. Then:

$$\mathbf{p} = qa\hat{\mathbf{z}} \tag{1.2.23}$$

Let this observation point be described by spherical coordinates (r, θ, ϕ), where θ describes the angle measured from the positive dipole axis.

We show the following

1. **Potential field**

   We find the distances of the observation points to the positive and negative charges. Let r_+ be the distance of the observation point from the positive charge, and r_- be its distance from the negative charge. We evaluate from the origin's perspective. From the cosine law, we have

$$r_+ = \sqrt{r^2 + \frac{a^2}{4} - ar\cos\theta} \tag{1.2.24}$$

$$r_- = \sqrt{r^2 + \frac{a^2}{4} + ar\cos\theta} \tag{1.2.25}$$

   Therefore, by superposition

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{q}{r_+} + \frac{-q}{r_-}\right) \tag{1.2.26}$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{r_+} - \frac{1}{r_-}\right) \tag{1.2.27}$$

or, explicitly,

$$V = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{\sqrt{r^2+\frac{a^2}{4}-ar\cos\theta}} - \frac{1}{\sqrt{r^2+\frac{a^2}{4}+ar\cos\theta}}\right) \tag{1.2.28}$$

2. **Electric field**

   The electric field then follows by

$$\mathbf{E} = -\nabla V \tag{1.2.29}$$

However, taking that directly from the potential seems mathematically cumbersome and tiring, so I have elected to not do it. Instead, I will derive the vector field for each point individually, and sum them at the end.

   (a) $+q$ **Electric field**

   We start with the assumption that we have a point source at the origin. We know its electric field

$$\mathbf{E}_+ = \frac{q}{4\pi\varepsilon_0}\frac{\hat{\mathbf{r}}}{r^2} = \frac{q}{4\pi\varepsilon_0}\frac{\mathbf{r}}{r^3} \tag{1.2.30}$$

   We then *translate* this point source $a/2$ units on the $+z$ direction.

$$\mathbf{E}_+ = \frac{q}{4\pi\varepsilon_0}\frac{(\mathbf{r}-\frac{a}{2}\hat{\mathbf{z}})}{r_+^3} \tag{1.2.31}$$

   (b) $-q$ **Electric field** We take the same logic from the previous step but translate the negative point down by $a/2$, to the $-z$ direction.

$$\mathbf{E}_- = -\frac{q}{4\pi\varepsilon_0}\frac{(\mathbf{r}+\frac{a}{2}\hat{\mathbf{z}})}{r_-^3} \tag{1.2.32}$$

Taking the vector sum of this results to the total electric field

$$\mathbf{E} = \mathbf{E}_+ + \mathbf{E}_-$$

Explicitly, we have

$$\mathbf{E} = \frac{q}{4\pi\varepsilon_0}\left(\frac{\mathbf{r}-\frac{a}{2}\hat{\mathbf{z}}}{r_+^3} - \frac{\mathbf{r}+\frac{a}{2}\hat{\mathbf{z}}}{r_-^3}\right) \tag{1.2.33}$$

3. **Point dipole limit** Now, we take the the limit as

$$a \to 0 \qquad\qquad q \to \infty \tag{1.2.34}$$

while keeping

$$p = qa \tag{1.2.35}$$

finite.

With $a \ll r$, we can take the binomial expansion of the inverse distances.

$$\frac{1}{r_+} \approx \frac{1}{r} + \frac{a\cos\theta}{2r^2} \tag{1.2.36}$$

$$\frac{1}{r_-} \approx \frac{1}{r} - \frac{a\cos\theta}{2r^2} \tag{1.2.37}$$

The difference of these is

$$\frac{1}{r_+} - \frac{1}{r_-} \approx \frac{a\cos\theta}{r^2} \tag{1.2.38}$$

We can then substitute this value for the $\frac{1}{r}$ in the potential field.

4. **Far field electric potential**

The potential field is defined as

$$V = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \tag{1.2.39}$$

We know that $\dfrac{1}{r} \to \dfrac{a\cos\theta}{r^2}$ at the limit. Therefore:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{qa\cos\theta}{r^2} \tag{1.2.40}$$

Since $qa = p$,

$$V = \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \tag{1.2.41}$$

At the dipole limit. In vector form this is:

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2} \tag{1.2.42}$$

or alternatively

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\mathbf{p}\cdot\mathbf{r}}{r^3} \tag{1.2.43}$$

5. **Far field electric field**

We take the negative of the gradient of (1.2.41)

$$\mathbf{E} = -\nabla V \tag{1.2.44}$$

Expressing this explicitly (in spherical basis vectors), we have

$$\mathbf{E} = \begin{bmatrix} \dfrac{\partial}{\partial r} \\ \dfrac{1}{r}\dfrac{\partial}{\partial \theta} \\ \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial \phi} \end{bmatrix} \frac{1}{4\pi\varepsilon_0}\frac{p\cos\theta}{r^2} \tag{1.2.45}$$

Evaluating this, we have

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0} \begin{bmatrix} \dfrac{\partial}{\partial r} \\ \dfrac{1}{r}\dfrac{\partial}{\partial \theta} \\ \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial \phi} \end{bmatrix} \frac{p\cos\theta}{r^2} \tag{1.2.46}$$

$$= \frac{1}{4\pi\varepsilon_0} \begin{bmatrix} \dfrac{\partial}{\partial r}\dfrac{p\cos\theta}{r^2} \\ \dfrac{1}{r}\dfrac{\partial}{\partial \theta}\dfrac{p\cos\theta}{r^2} \\ \dfrac{1}{r\sin\theta}\dfrac{\partial}{\partial \phi}\dfrac{p\cos\theta}{r^2} \end{bmatrix} \tag{1.2.47}$$

$$= \frac{1}{4\pi\varepsilon_0} \begin{bmatrix} -\dfrac{2p\cos\theta}{r^3} \\ -\dfrac{p\sin\theta}{r^3} \\ 0 \end{bmatrix} \tag{1.2.48}$$

$$\tag{1.2.49}$$

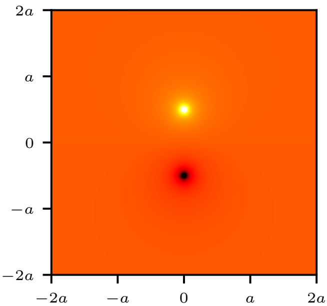
Hence, in the spherical basis $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}}, \hat{\boldsymbol{\phi}})$, the electric field has the form

$$\mathbf{E} = \frac{p}{4\pi\varepsilon_0 r^3} \begin{bmatrix} 2\cos\theta \\ \sin\theta \\ 0 \end{bmatrix} \tag{1.2.50}$$

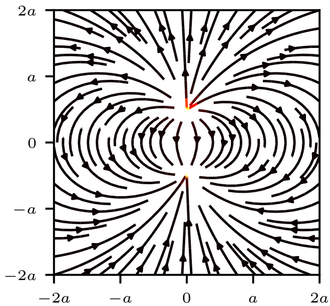
In vector form, it has the expression

$$\mathbf{E}(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^3}(3(\mathbf{p}\cdot\hat{\mathbf{r}})\hat{\mathbf{r}} - \mathbf{p}) \qquad (1.2.51)$$

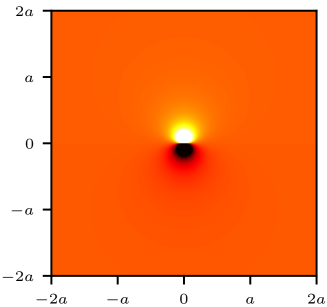
Here are the plots of V and $\mathbf{E}$ for each case solved in the problem.



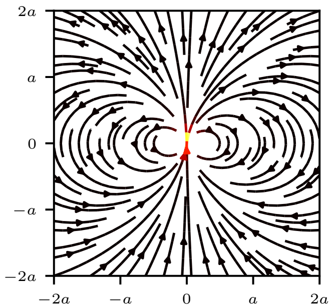
(a) Dipole potential V



(b) Dipole vector field $\mathbf{E}$



(c) Dipole limit potential V



(d) Dipile limit vector field $\mathbf{E}$

Figure 1.9: Dipole electric potential fields at distance a and at the limit $a \to 0$

1.3 Electric Potential

1.3.1 Introduction

The electric field $\mathbf{E}$ is not just *any* old vector function. It is a special kind of vector function, one whose curl is zero. $\mathbf{E} = y\hat{\mathbf{x}}$, for example, could not possibly be an electrostatic field. No set of charges, regardless of their sizes and positions, could ever produce such a field.

We're going to exploit this special property of electric fields to reduce a vector problem (finding $\mathbf{E}$), to a much simpler *scalar* problem. Theorem 1.4 states that any vector field whose curl is zero is equal to the gradient of some scalar field. What we are going to do now amounts to a proof of such claim, in the context of electrostatics.

Because $\nabla \times \mathbf{E} = \mathbf{0}$, the line integral of $\mathbf{E}$ around any closed loop is zero.[5] Because $\oint \mathbf{E}\cdot d\boldsymbol{\ell} = 0$, the line integral of $\mathbf{E}$ from a point $\mathbf{a}$ to point $\mathbf{b}$, is the same for all paths[6]. Because the line integral is independent of path, we can define a function

$$V(\mathbf{r}) \equiv -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell} \tag{1.3.1}$$

Here, $\mathcal{O}$ is some standard reference point on which we have agreed to beforehand. V then depends only on the point $\mathbf{r}$. It is called the **Electric potential**.

The potential *difference* between two points $\mathbf{a}$ and $\mathbf{b}$ is:

$$\begin{aligned} V(\mathbf{b}) - V(\mathbf{a}) &= -\int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} + \int_{\mathcal{O}}^{\mathbf{a}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= -\int_{\mathcal{O}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} - \int_{\mathbf{a}}^{\mathcal{O}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} \end{aligned}$$

Now, the fundamental theorem for gradients state that

$$V(\mathbf{b}) - V(\mathbf{a}) = \int_{\mathbf{a}}^{\mathbf{b}} (\nabla V) \cdot d\boldsymbol{\ell} \tag{1.3.2}$$

[5] From Stokes' Theorem

[6] otherwise you could go *out* along a path a and return along a path b, and obtain $\oint \mathbf{E} \cdot d\boldsymbol{\ell} \neq 0$

so,

$$\int_{\mathbf{a}}^{\mathbf{b}} (\nabla V)\cdot d\boldsymbol{\ell} = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E}\cdot d\boldsymbol{\ell} \tag{1.3.3}$$

Since, finally, this is true for any points **a** and **b**, the integrands must be equal. Hence:

Theorem 1.10 : Electric fields come from scalar potentials

$$\mathbf{E} = -\nabla V \tag{1.3.4}$$

Equation (1.3.4) is the differential version of (1.3.1); it says that the electric field is the gradient of a scalar potential, which we set out to prove.

Notice the subtle but crucial role played by the path independence (or equivalently, the fact that $\nabla\times\mathbf{E} = \mathbf{0}$) in this argument. If the line integral of **E** depended on the path taken, then the definition of V (1.3.4), would become nonsense. It simply would not define a function, since changing the path would alter the value of $V(\mathbf{r})$.

1.3.2 Some comments on Potentials

1.3.2.1 The name

The word "potential" is sort of a misnomer, because it reminds you of the concept of potential *energy*. While there is a connection between the two, that is not precisely what electric potential means, and they are completely different terms. Although, the name potential here comes from the fact that taking the gradient results in a force, which is analogous to gravitational potential Φ,

$$\mathbf{F} = -\nabla\Phi,$$

and in fact all kinds of other *potentials* used in other physics terms. Hence, for this reference, the name potential is appropriate. Incidentally, a surface over which the potential is constant is called an **equipotential**

1.3.2.2 Advantage of the potential formula

Knowledge of the potential field V immediately means that you can get **E**, just by takign the gradient $\mathbf{E} = -\nabla V$. This is quite fascinating, since $\mathbf{E}$ is a vector quantity of three components, while V is simply a scalar of one component. As such, the possibility that one function contains all the information of three independent functions is quite absurd. However, the answer is that the three components of $\mathbf{E}$ are not really quite independent. They are, in fact, explicitly related by the condition we used to prove the potential, and that is $\nabla \times \mathbf{E} = \mathbf{0}$

In terms of components, they can be expressed as

$$\frac{\partial E_x}{\partial y} = \frac{\partial E_y}{\partial x} \qquad \frac{\partial E_z}{\partial y} = \frac{\partial E_y}{\partial z} \qquad \frac{\partial E_x}{\partial z} = \frac{\partial E_z}{\partial x}$$

As such, the components of $\mathbf{E}$ are really coupled differential equations that relate to each other. This brings us back to our observation in the beginning, $\mathbf{E}$ is a *special kind* of vector field, and the potential formulation exploits these feauture to a maximun advantage, reducing a vector problem to a scalar one, where there is no need to inconvenience ourselves with components.

1.3.2.3 The arbitrary reference point

The choice for an arbitrary reference point $\mathcal{O}$ seems quite strange. However, here, we argue that the electric potential is inherently defined only up to an arbitrary additive constant, because the choice of reference point $\mathcal{O}$ is arbitrary. If the reference point is changed from, say $\mathcal{O}$ to $\mathcal{O}'$, then

$$\begin{aligned} V'(r) &= -\int_{\mathcal{O}'}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E} \cdot d\boldsymbol{\ell} - \int_{\mathcal{O}'}^{\mathcal{O}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= V(\mathbf{r}) + K \end{aligned}$$

where K is a constant independent of $\mathbf{r}$. Consequently,

$$V'(\mathbf{b}) - V'(\mathbf{a}) = V(\mathbf{b}) - V(\mathbf{a})$$

and as such, potential differences are physically meaningful, while the absolute value of the potential is not. Since $V'(\mathbf{r})$ is only $V(\mathbf{r}) + K$, then

$$\nabla V' = \nabla V$$

and as such the electric field is likewise unchanged even though the absolute potential is different. The choice of reference point is therefore mostly a matter of convenience. One can choose $V = 0$ at any convenient location, much like choosing where to define zero altitude. What matters physically is the difference in potential between two points. Therefore, saying a point has a particular *absolute* potential has no intrinsic meaning unless a reference has been specified.

As such, there is no physically privileged zero of electric potential, and one may choose the reference point wherever convenient, because changing it does not alter the potential difference nor the vector field that arises from the potential.

There is, however, a particularly convenient convention in electrostatics: take the potential to vanish at infinity, i.e.,

$$V(\infty) = 0$$

Ordinarily, then, we "set the zero of potential at infinity". Since $V(\mathcal{O}) = 0$, choosing a reference point is equivalent to selecting a place where V is to be zero). This works naturally for localized charge distributions because the electric field becomes negligible sufficiently far away. For an infinite plane of charge, however, this convention fails: the potential does not approach a finite value at infinity. For example,

$$V(z) = -\int_{\infty}^{z} \frac{\sigma}{2\varepsilon_0}\, dz = -\frac{\sigma}{2\varepsilon_0}(z - \infty)$$

The potential in this case is not finite. The remedy is simply to choose some other reference point. However, this problem is merely theoretical. In practice, there is no electric potential that extends infinitely, and one can always define a point be $V = 0$.

1.3.2.4 Potentials obey superposition

The original superposition principle pertains to the force acting on some test charge q. It says that the total force acting on q is the vector sum of all the forces attributable to the source charges individually.

$$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \cdots$$

Dividing by q, we see that the electric field also obeys the principle of superposition.

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \cdots$$

If we integrate from the common reference point to $\mathbf{r}$, it follows that the potential also satisfies such principle

$$V = V_1 + V_2 + \cdots$$

Therefore, the potential at any given point is the sum of the potentials due to all the charges separately. Only this time, the quantity is an ordinary sum, not a vector sum, which makes it more convenient and easier to work with.

1.3.2.5 Units of Potential

In the customary *Système International* (SI) units, force is measured in newtons and charge in coulombs, and as such, electric fields are measured in newtons per coulomb. Accordingly, potentials are measured in newton-meters per coulomb.

A joule per coulomb is called a **volt**

$$1\,\mathrm{V} = 1\frac{\mathrm{Nm}}{\mathrm{C}}$$

1.3.3 Poisson's and Laplace's Equation

We found in §1.3.1 that the electric field can be written as the gradient of a scalar potential

$$\mathbf{E} = -\nabla V$$

The question then arises: What do the divergence and curl of the electrice field $\mathbf{E}$, $\nabla \cdot \mathbf{E} = \rho/\varepsilon_0$ and $\nabla \times \mathbf{E} = \mathbf{0}$, look like in terms of V? Well, we have

$$\begin{aligned}
\mathbf{E} &= -\nabla V \\
\nabla \cdot \mathbf{E} &= \nabla \cdot (-\nabla V) \\
\nabla \cdot \mathbf{E} &= -\nabla^2 V
\end{aligned}$$

And so, apart from the persistent minus sign, the divergence of $\mathbf{E}$ is the Laplacian of V. Gauss's Law, then, says that

Theorem 1.11 : Poisson's Equation

$$\nabla^2 V = -\frac{\rho}{\varepsilon_0} \tag{1.3.5}$$

This is known as **Poisson's equation**. In regions where there are no charge, so $\rho = 0$, Poisson's equation reduces to **Laplace's equation**

Theorem 1.12 : Laplace's Equation

$$\nabla^2 V = 0 \tag{1.3.6}$$

For the curl, we have

$$\nabla \times \mathbf{E} = \nabla \times (-\nabla V) = \mathbf{0}$$

But that is no condition on V. The curl of a gradient is *always* zero. In fact, we used the curl law to show that $\mathbf{E}$ could be expressed as a gradient of a scalar, so it's not really surprising that this is the case. $\nabla \times \mathbf{E} = \mathbf{0}$ *permits* $\mathbf{E} = -\nabla V$, and in return, $\mathbf{E} = -\nabla V$ *guarantees* $\nabla \times \mathbf{E} = \mathbf{0}$. It only takes one differential equation, Poisson's equation, to determine V, because V is a scalar. For $\mathbf{E}$, we needed two, the divergence and the curl.

In this case, then, we can say that $\nabla^2 V = -\dfrac{\rho}{\varepsilon_0}$ is the ***field equation*** governing electrostatics.

1.3.4 The Potential of a Localized Charge Distribution

We defined V in terms of $\mathbf{E}$, and found that it is more convenient to do so since it makes calculations significantly easier. Normally, though, it's $\mathbf{E}$ that we are looking for (if we already knew $\mathbf{E}$, there wouldn't be much point in calculating V). The idea is that it might be easier to get V first, and then calculate $\mathbf{E}$ by taking the gradient. Typically, then, we know where the charge is (i.e, we know ρ), and we want to find V. Poisson's equation relates V and ρ, but unfortunately, it is formulated in such a way that it provides ρ given that we already knew V, whereas we want to know V, knowing ρ. What we must do then, is "invert" Poisson's equation. For this section, that is the task which we set out to do. However, we shall do it by roundabout means, beginning, as usual, with a point charge at the origin.

The electric field is

$$\mathbf{E} = \frac{1}{4\pi\varepsilon_0}\frac{1}{r^2}\hat{\mathbf{r}}$$

and the line element is

$$d\boldsymbol{\ell} = dr\,\hat{\mathbf{r}} + r\,d\theta\,\hat{\boldsymbol{\theta}} + r\sin\theta\,d\phi\,\hat{\boldsymbol{\phi}}$$

Hence, we have

$$\mathbf{E}\cdot d\boldsymbol{\ell} = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}\,dr$$

Setting the reference point at infinity, the potential of the point charge q at the origin is

$$\begin{aligned} V(r) &= -\int_{\mathcal{O}}^{r} \mathbf{E}\cdot d\boldsymbol{\ell} \\ &= -\frac{1}{4\pi\varepsilon_0}\int_{\infty}^{r}\frac{q}{r'^2}\,dr' \\ &= \frac{1}{4\pi\varepsilon_0}\frac{q}{r'}\bigg|_{\infty}^{r} \\ &= \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \end{aligned}$$

We see here the advantage of using infinity as a reference point; it kills the lower limit of the integral. Notice the sign of V; presumably, the conventional minus sign in the definition (1.3.5) was chosen in order to make the potential of a positive charge come out positive. It is useful to remember that regions og positive charge are potential 'hills', whereas regions of negative charge are potential 'valleys', and the electric field always points *downhill* from the high point of the potential to the low point.

In general, the potential of a point charge q is

$$V(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0}\frac{q}{r} \tag{1.3.7}$$

where r is the distance from q to a point $\mathbf{x}$[7]. i.e., $\mathbf{r} = \mathbf{x} - \mathbf{x}_q$, where $\mathbf{x}_q$ is the position of q. Invoking the superposition principle, then, we find

[7] Forgive the change of variables, the text becomes confusing so this is to help me

that the potential of a collection of charges is

$$V(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \sum_i \frac{q}{r_i}, \tag{1.3.8}$$

or, for a continuous distribution,

$$V(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{1}{r}\, dq \tag{1.3.9}$$

In particular, for a volume charge, it is,

Theorem 1.13 : Potential for a charge distribution

$$V(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{x}')}{r}\, dV \tag{1.3.10}$$

Here, $\mathbf{x}'$ denotes the test point being integrated over all $\mathbb{R}^3$ in the integral. This is the equation we were looking for, which tells us how to compute V when we know ρ. It is, if you will, the *solution* to Poisson's equation, for a localized charge distribution. Compare this to the corresponding equation for the electric field:

$$\mathbf{E}(\mathbf{x}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\rho(\mathbf{x}')}{r^2} \hat{\mathbf{r}}\, dV \tag{1.3.11}$$

The main point here is that the pesky unit vector $\hat{\mathbf{r}}$ is gone, so there is no need to fuss with components. Here, we go back to denoting general position to $\mathbf{r}$, since it is an arbitrary point and may be relabeled.. The potentials of line and surface charges are

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\lambda(\mathbf{r}')}{r^2}\, d\ell \tag{1.3.12}$$

and

$$V(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \int \frac{\sigma(\mathbf{r}')}{r^2}\, dS \tag{1.3.13}$$

We warn you, however, that everything presented in this section is predicated upon the assumption that the reference point is at infinity. This is hardly apparent from (1.3.10), but remember that we *got* that equation of the potential from a point charge at the origin, which is only valid when $\mathcal{O} = \infty$. If one tries to apply these formulae to one of those artificial problems where the charge itself extends to infinity, the integral will diverge.

1.3.5 Boundary Conditions

In the typical electrostatic problem, one is given a source charge distribution ρ, and is tasked to fined the electric field $\mathbf{E}$ it produces. Unless the symmetry of the problem allows a solution by Gauss's law, it is generally in one's advantage to calculate the potential first, as an intermediate step. These are the three fundamental quantities in electrostatics are: $\rho, \mathbf{E}$, and V. We have, in the course of the discussion, derived most of the formulae interrelating them. We began with just two experimental observations: (1) the principle of superposition–a broad general rule applying to *all* electromagnetic forces, and (2) Coulomb's law–the fundamental law of electrostatics. From these, all else followed.

One may notice, while testing out these problems, that the electric field always undergoes a discontinuity when crossing a surface charge σ. It is, in fact, a simple matter to find the *amount* by which $\mathbf{E}$ changes at such a boundary. Suppose we draw a wafer-thin Gaussian pillbox, extending just barely over the edge in each direction (Fig. 1.10), Gauss's law says that

$$\oint_S \mathbf{E} \cdot d\mathbf{S} = \frac{1}{\varepsilon_0} Q_{\text{enc}} = \frac{1}{\varepsilon_0} \sigma A$$

where A is the area of the pillbox lid (If σ varies from point to point or the surface is curved, we must pick A to be extremely small.) Now, the *sides* of the pillbox contribute nothing to the flux. In the limit, as the thickness ε goes to zero, so we are left with

$$E^{\perp}_{\text{above}} - E^{\perp}_{\text{below}} - = \frac{1}{\varepsilon_0} \sigma \tag{1.3.14}$$

where $E^{\perp}_{\text{aboce}}$ denotes the component of $\mathbf{E}$ that is perpendicular to the surface immediately above, and E_{below} is the same, only below the surface. For consistency, we let *upward* be the positive direction for both.

A conclusion we can form is that the normal component of $\mathbf{E}$ is discontinuos by an amount σ/ε_0 at any boundary. In particular, when there is *no* surface charge, $E^{\perp}$ is continuous, as for instance, at at the surface of a uniformly charged solid sphere.

The *tangential* component of $\mathbf{E}$, by contrast, is *always* continuous. For if we apply Eq. 1.2.20,

$$\oint \mathbf{E} \cdot d\boldsymbol{\ell} = 0 \tag{1.3.15}$$

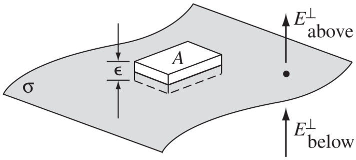


Figure 1.10: The setup of the Gaussian pillbox problem

to the thin rectangular loop of Fig. TBA, the ends give nothing (as $\varepsilon \to 0$), and both sides give $E^{||}_{\text{above}} l - E^{||}_{\text{below}} l$, so

$$\mathbf{E}^{||}_{\text{above}} = \mathbf{E}^{||}_{\text{below}} \tag{1.3.16}$$

where $\mathbf{E}^{||}$ stands for the components of $\mathbf{E}$ *parallel* to the surface. The boundary conditions on $\mathbf{E}$, Fig. TBA and Fig. TBA, can be combined into a single formula:

$$\mathbf{E}_{\text{above}} - \mathbf{E}_{below} = \frac{\sigma}{\varepsilon_0}\hat{\mathbf{n}} \tag{1.3.17}$$

where $\hat{\mathbf{n}}$ is a unit vector perpendicular to the surface, pointing from *below* to *above*.

The potential, meanwile, is continuos across *any* boundary (Fig. TBA), since

$$V_{\text{above}} - V_{\text{below}} = -\int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} \tag{1.3.18}$$

as the path length shrinks to zero, so too does the integral:

$$V_{\text{above}} = V_{\text{below}} \tag{1.3.19}$$

However, the *gradient* of V inherits the discontinuity $\mathbf{E}$; since $\mathbf{E} = -\nabla V$. Eq. 1.3.17 implies that

$$\nabla V_{\text{above}} - \nabla V_{below} = -\frac{\sigma}{\varepsilon_0}\hat{\mathbf{n}} \tag{1.3.20}$$

or, mroe conveniently,

$$\frac{\partial V_{\text{above}}}{\partial n} - \frac{\partial V_{\text{below}}}{\partial n} = -\frac{\sigma}{\varepsilon_0} \tag{1.3.21}$$

where

$$\frac{\partial V}{\partial n} = \nabla V \cdot \hat{\mathbf{n}} \tag{1.3.22}$$

denotes the **normal derivative** of V, i.e., the rate of change in the direction perpendicular to the surface.

Note that these boundary conditions relate the fields and potentials *just* above and *just* below the surface. For example, the derivatives in (1.3.21) are the *limiting* values as we approach the surface from either side.

1.4 Work and Energy

1.4.1 The Work it takes to Move a Charge

Suppose you have a stationary configuration of source changes, and you want to move a test charge Q from point $\mathbf{a}$ to point $\mathbf{b}$. A question can be raised: How much work will you have to do? At any point along the path, the electric force on Q is $\mathbf{F} = Q\mathbf{E}$. The force *you* must exert, in opposition to this electrical force, is $-Q\mathbf{E}$. The work done is therefore

$$\begin{aligned} W &= \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{F} \cdot d\boldsymbol{\ell} \\ &= -Q \int_{\mathbf{a}}^{\mathbf{b}} \mathbf{E} \cdot d\boldsymbol{\ell} \\ &= Q[V(\mathbf{b}) - V\mathbf{a}] \end{aligned}$$

Notice that the answer is indepenent of the path you take from $\mathbf{a}$ to $\mathbf{b}$; in mechanics, then, we would call the electrostatic force *conservative.* Dividing through by Q, we have

$$V(\mathbf{b}) - V(\mathbf{a}) = \frac{W}{Q} \tag{1.4.1}$$

In other words, the potential difference between points $\mathbf{a}$ and $\mathbf{b}$ is equal to the work for unit charge required to carry a particle from $\mathbf{a}$ to $\mathbf{b}$. In particular, if you want to bring Q in from far away and stick it at a point $\mathbf{r}$, the work you must do is

$$W = Q[V(\mathbf{r}) - V(\infty)]$$

and so, if you have set the referece point at infinity, the work done is

Theorem 1.14 : Work done in an electric potential

$$W = QV(\mathbf{r}) \tag{1.4.2}$$

In that sense, *potential* is the potential *energy* (the work it takes to create the system) *per unit charge* (just as the *field* is the *force* per unit charge).

1.4.2 The Energy of a Point Charge Distribution

How much work would it take to assemble an entire *collection* of point charges? Imagine bringing in the charges, one by one, from far away. The first charge q_1, takes *no* work, since there is no field yet to fight against. Now, we bring in q_2. Acording to (1.4.2), this will cost $q_1 V_1(\mathbf{r}_2)$, where V_1 is the potential due to q_1, and $\mathbf{r}_2$ is the place that we're putting q_2:

$$W_2 = \frac{1}{4\pi\varepsilon_0} q_2 \left(\frac{q_1}{r_{12}} \right)$$

where r_{12} is the distance between q_1 and q_2 once they are position. As you bring in each charge, pin it down to its final location, so that it does not move when you bring the next charge. Now, we bring q_3, this requires work $q_3 V_{1,2}(\mathbf{r}_3)$, where $V_{1,2}$ is the potential due to charges q_1 and q_2. Thus

$$W_3 = \frac{1}{4\pi\varepsilon_0} q_3 \left(\frac{q_1}{r_{13}} + \frac{q_2}{r_{23}} \right)$$

Similarly, the extra work to bring in a fourth charge q_4 is

$$W_4 = \frac{1}{4\pi\varepsilon_0} q_4 \left(\frac{q_1}{r_{14}} + \frac{q_2}{r_{24}} + \frac{q_3}{r_{34}} \right)$$

The *total* work necessary to assemble the first four charges, then, is

$$W = \frac{1}{4\pi\varepsilon_0} \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_1 q_4}{r_{14}} + \frac{q_2 q_3}{r_{23}} + \frac{q_2 q_4}{r_{24}} + \frac{q_3 q_4}{r_{34}} \right)$$

Therefore, in general

$$W = \frac{1}{4\pi\varepsilon_0} \sum_{i=1}^{n} \sum_{j>i}^{n} \frac{q_i q_j}{r_{ij}} \tag{1.4.3}$$

The stipulation $j > i$ is to remind us not to count the same pair twice. A neater way to accomplish this is *intentionally* to count each pair twice, and then halve it:

$$W = \frac{1}{8\pi\varepsilon_0} \sum_{i=1}^{n} \sum_{j\neq i}^{n} \frac{q_i q_j}{r_{ij}} \tag{1.4.4}$$

We must still avoid $i = j$, of course. Notice that in this form, the answer plainly does not depend on the order in which you assemble the charges, since every pair occurs in the sum.

Pulling out the facter q, we have

$$W = \frac{1}{2}\sum_{i=1}^{n} q_i \left(\sum_{i \neq j}^{n} \frac{1}{4\pi\varepsilon_0} \frac{q_j}{r_{ij}} \right)$$

The term inside parenthesis is the potential at point $\mathbf{r}_i$ (the position of q_i) due to all the *other* charges, and we mean all of them now, not just the ones that were present at some stage during the assembly. Thus

$$W = \frac{1}{2}\sum_{i=1}^{n} q_i V(\mathbf{r}_i) \tag{1.4.5}$$

That's how much work it takes to assemble a configuration of point chages; it is also the same amount of work returned if the system is dismantled. In the meantime, it represents energy stored in the configuration.

1.4.3 The Energy of a Continuous Charge Distribution

For a volumme charge density ρ, (1.4.5) becomes

$$W = \frac{1}{2}\int \rho V \, d\tau \tag{1.4.6}$$

(We use $d\tau$ for the volume element dV, because V is used as potential, additionally, the corresponding integrals for line and surface charges would be $\int \lambda V \, d\ell$ and $\int \sigma V \, dS$). There is a neat way to write this result, in which ρ and V are eliminated in favor of $\mathbf{E}$. First, we use Gauss's law to express ρ in terms of $\mathbf{E}$.

$$\rho = \varepsilon_0 \nabla \cdot \mathbf{E}$$

so,

$$W = \frac{\varepsilon_0}{2}\int (\nabla \cdot \mathbf{E}) V \, d\tau$$

Now, we use integration by parts to transfer the derivative from $\mathbf{E}$ to V,

$$W = \frac{\varepsilon_0}{2}\left(-\int \mathbf{E} \cdot (\nabla V) \, d\tau + \oint V \mathbf{E} \cdot d\mathbf{S} \right)$$

But $\nabla V = -\mathbf{E}$, so

$$W = \frac{\varepsilon_0}{2}\left(\int_{\mathcal{V}} E^2\, d\tau + \oint V\mathbf{E}\cdot d\mathbf{S}\right) \tag{1.4.7}$$

But what volume $\mathcal{V}$ *is* this over which we are integrating? Let us return to the expression with which we began, Eq. (1.4.6). From its derivation, it is clear that the volume need only encompass the region containing the charge. However, any *larger* volume is equally valid: the additional region contributes nothing to the integral, since $\rho = 0$ there.

With this in mind, consider Eq. (1.4.8). What happens as the volume is enlarged beyond the minimum required to enclose all the charge? The volume integral of E^2 can only increase, since its integrand is positive. Consequently, the surface integral must decrease by a corresponding amount in order for the total to remain unchanged. At sufficiently large distances from a localized charge distribution, E falls off as $1/r^2$ and V as $1/r$, while the surface area grows as r^2. Thus, the surface integral falls off approximately as $1/r$.

It is important to note that Eq. (1.4.6) yields the same total energy W for any volume enclosing all the charge. What changes with the choice of volume is the division of this energy between the volume and surface integrals: as the volume is increased, the contribution from the volume integral increases while that from the surface integral decreases. In the limiting case, the volume may be extended over all space. The surface integral then vanishes, leaving, for all space

$$W = \frac{\varepsilon_0}{2}\int E^2\, d\tau \tag{1.4.8}$$

1.4.4 Comments on Electrostatic Energy

1.4.4.1 Observed Inconsistencies

Equation (1.4.8) appears to imply that the electrostatic energy of a stationary charge distribution is necessarily positive. This seems inconsistent with Eq. (1.4.5), from which Eq. (1.4.8) is derived, since the latter may yield either a positive or negative value. For example, for two equal and opposite point charges separated by a distance r, Eq. (1.4.5) gives

$$W = -\frac{1}{4\pi\varepsilon_0}\frac{q^2}{r}.$$

The apparent contradiction raises two questions: what accounts for the discrepancy, and which expression represents the electrostatic energy?

Both expressions are correct, but they refer to different definitions of the energy being considered. Equation (1.4.5) gives the work required to assemble a specified collection of point charges, beginning with the charges already formed. It therefore accounts only for the interaction energy between distinct charges. It does not include the energy required to create the individual point charges themselves.

Equation (1.4.8), by contrast, represents the total energy stored in the electric field of a charge distribution. For a point charge, this expression includes the charge's self-energy, which is divergent. For a point charge q,

$$W = \frac{\varepsilon_0}{2} \int E^2 \, d\tau = \frac{\varepsilon_0}{2(4\pi\varepsilon_0)^2} \int_0^\infty \frac{q^2}{r^4} \left(r^2 \sin\theta \, dr \, d\theta \, d\phi\right),$$

and hence

$$W = \frac{q^2}{8\pi\varepsilon_0} \int_0^\infty \frac{1}{r^2} \, dr = \infty.$$

Thus, Eq. (1.4.8) contains not only the interaction energy between charges, but also their individual self-energies. For ideal point charges, these self-energies are infinite. Equation (1.4.5) is consequently more useful for systems of point charges when only the physically relevant interaction energy is required. In such applications, the point charges are treated as already existing objects, and only the work associated with changing their relative configuration is considered.

The apparent inconsistency can be traced to the transition between Eqs. (1.4.5) and (1.4.6). In Eq. (1.4.5), the potential evaluated at the position $\mathbf{r}_i$ is the potential produced by all charges *other than* q_i. In Eq. (1.4.8), however, $V(\mathbf{r})$ denotes the full electrostatic potential, including the contribution associated with the charge at $\mathbf{r}$ itself.

For a continuous charge distribution, this distinction does not arise in the same way. The charge contained in an infinitesimal neighborhood of any particular point is vanishingly small, so its individual contribution to the potential does not produce a separate self-energy term of the type encountered for an ideal point charge. Consequently, the field-energy expression may be applied directly to a continuous distribution.

For discrete point charges, however, the self-energy must be distinguished from the interaction energy. Equation (1.4.5) should therefore be interpreted as the interaction energy of the point-charge configuration, whereas Eq. (1.4.8) gives the total field energy, including the self-energy contributions.

1.4.4.2 Where is the energy stored?

Eqs. (1.4.6) and (1.4.8) offer two different ways of calculating the same thing. The first is an integral over the charge distribution; the second is an integral over the field. This can involve completely different regions. For instance, in the case of the spherical shell, the charge is confied to the surface, whereas electric field is everywhere *outside* this surface. Where is the energy? Is it stored in the field, as (1.4.8) seems to suggest, or is it stored in the charge, as (1.4.6) implies. At present, this is an unanswerabe question. We can calculate the total energy is, and multiple ways to calculate it, yet it is impertinent to worry about *where* the energy is located.

In the context of radiation theory, it is useful, and in general relativity, it is essential, to regard the energy stored in the field, with a density

$$\frac{\varepsilon_0}{2}E^2 = \text{energy per unit volume.}$$

But in electronics, one could just as well say that it is stored in the charge, with density $\frac{1}{2}\rho V$. The differences is purely a matter of bookkeeping.

1.4.4.3 The superposition principle

Because electrostatic energy is *quadratic* in the fields, it does *not* obey a superposition principle. The energy of a compound system is *not* the sum of the energies of its parts considered separately–there are also "cross terms"

$$
\begin{aligned}
W_{\text{tot}} &= \frac{\varepsilon_0}{2} \int E^2 \, d\tau \\
&= \frac{\varepsilon_0}{2} \int (\mathbf{E}_1 + \mathbf{E}_2)^2 \, d\tau \\
&= \frac{\varepsilon_0}{2} \int E_1^2 + E_2^2 + 2\mathbf{E}_1 \cdot \mathbf{E}_2 \, d\tau \\
&= W_1 + W_2 + \varepsilon_0 \int \mathbf{E}_1 \cdot \mathbf{E}_2 \, d\tau
\end{aligned}
$$

For example, if you double the charge everywhere, you *quadruple* the total energy.

1.5 Conductors

1.5.1 Basic Properties

1.5.2 Induced Charges

1.5.3 Surface Charge and Force on a Conductor

1.5.4 Capacitors

1.6 Calculating Scalar Potentials

1.7 Dielectrics

1.8 Electric Displacement

1.9 Energy in dielectrics, Magnetostatics

1.10 Magnetic Vector Potential

1.11 Unit Summary

Unit 2

Magnetostatics

2.1 Introduction

2.2 Magnetic fields in Matter

2.3 Boundary Value Problems

2.4 Unit Summary

Author's Note

This entire note section is just about me and my musings, about this subject, and about me. So, there isn't really much to learn here, this is simply my outlet regarding what I have experienced and what I have to say.

I took Physics 131 about two semesters ago, when I was in fourth year. That was supposed to be quite easy, Sir Paraan is quite lenient when it comes to subject outputs when compared to other subjects, that being only giving a single problem set, and a singe exam. I aced the problem set and the exam during that time, having very high marks on both. However, on the second requirement, that was where I was caught lacking.

I never quite liked note taking during lectures. I believe it is distracting to a learner, and a student learns best while actively listening to the lecturer as he speaks, and digest these thoughts and learnings after the class is done, much like intellectual exercise. As such, I never really took notes, not even in my highschool days, and simply learned everything through listening. This was how I worked for years, so taking notes was not in my nature as a student, and it worked for me, graduating high school distinguished, in fact.

The notebook was a pivotal part of the subject, however, accounting for over 50% of the grade, and so looking back I really should have focused on doing it, much like a project. However, due to my rather busy schedule that time, what with work and starting laboratory internships and other activities that took my time, I simply had no time to work on the notebooks. Looking back, I should have just treated it the same way as weekly problem sets and should have done my due diligence, but unfortunately, I was simply too swamped with workload that I couldn't really work on it. I began the work only about five days before tha deadline, and couldn't complete it on time for said deadline. Futhermore, during the second submission, I was sufficiently burned out that I really couldn't do much anymore. When the time that I have decided that I was not gonna pass this subject, the dropping deadline has lapsed, and so I was stuck. It was at that time that I regretted not doing the notebook.

Anyhow, grading period came, and I recieved a 5.0, which is quite tragic, yes. It set me back another sem, and I should be graduating right now. However, it has passed, and it is no use to regret from our past mistakes, and instead take it as learning opportunities.

I am in fifth year now, a year more than one should have taken in their undergraduate years. I'm academically old, about to graduate this year. I feel old, despite being twenty three. Despite my exemplary performance in my secondary education, I am significantly more laid back and exert the bare minimum effort during my college years. I am part of the pipeline of gifted students in their youth, turned academic disappointments during adulthood. All those years seem wasted, but yes, I am now about to graduate from this god forsaken University. It's time to get my shit together and get my act right.

So, for this semester, on account of having lesser workload in general, I have decided to go all out on the notebook, giving it my best effort, so that I may submit a great work when the times of submission comes, and I can be proud of it. Go big or go home, as they say. Hopefully, what comes out of this work, though simply note taking and rather derivative, is of acceptable merit to both Sir Paraan, or his checker, and a work which, while I am averse in doing, I can look back on with mild happiness, and be proud.

End.

PHYSICA VINCIT OMNIA